南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

生物医学工程系
Department of Biomedical Engineering

Medical Image Processing

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2026/9/18

Course Lecturer

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Education and Training





Home Lab members Research Publications News Join us

LI Yiming Google Scholar ResearcherID

Associate Professor
Department of Biomedical Engineering

Education

2010-2015 Ph. D., Biophysics, Karlsruhe Institute of Technology

2009-2010 M. Sc., Medical Physics, Heidelberg University

2005-2009 B.E., Biomedical Engineering, Shanghai Jiao Tong University

Employment

2019-present Department of Biomedical Engineering, Southern University of Science and Technology, Research Associate Scientist

2016-2019 European Molecular Biology Laboratory, Heidelberg, Postdoc

2016.02-2016.09 Yale University， Visiting Scholar



Yiming Li, PhD

Associate Professor of Biomedical Engineering

Advanced Microscopy Imaging Laboratory

工学院南楼729

What we do

Seeing is believing! We firmly believe that resolution determines the depth of the research. We are interested to use super-resolution microscopy to show the wonders of the microscopic world. To achieve this, we collaborate heavily with biologists and develop a range of interdisciplinary technologies including new optics theories, advanced algorithms, smart hardware design and imaging strategies for better resolution, deeper depth, and faster speed. We embrace the open science and hope that our work will promote both the advancement of science and technology.

https://li-lab-sustech.github.io/

Who are we



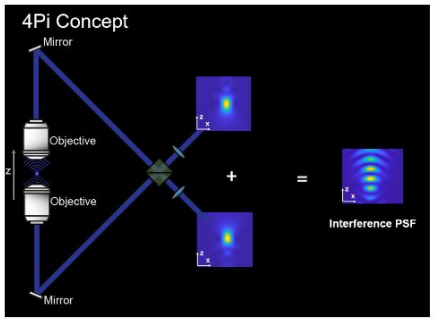
Our group is a mix of physicist, engineer, computer scientist, biologist and chemist.

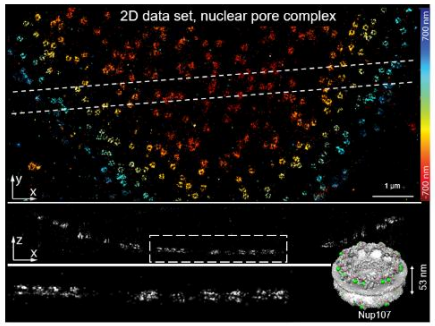


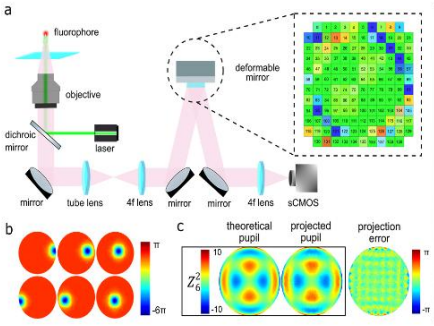
Yiming Li, PhD

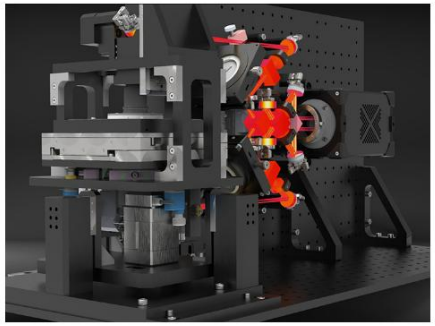
Associate Professor of Biomedical Engineering

My overall aim is to push the resolution limit of super resolution imaging down to <10 nm in all 3D, enabling structural studies of large dynamic protein assemblies inside intact cells.









➢ Optics

Super resolution, Adaptive optics, Deep learning, Optics design (Zemax), 4Pi

➢ Programming

CUDA C/C++, Python, MATLAB, LabView, Java

➢ Biological Imaging

Single molecule, Fluorescent labeling, Dyes, DNA-PAINT

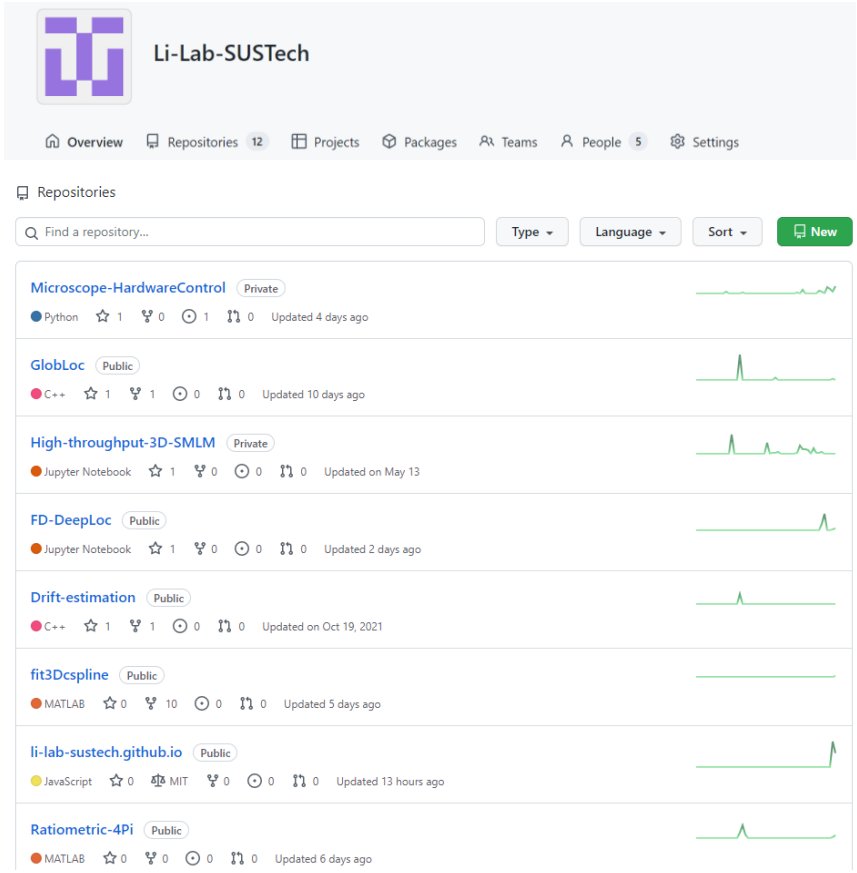


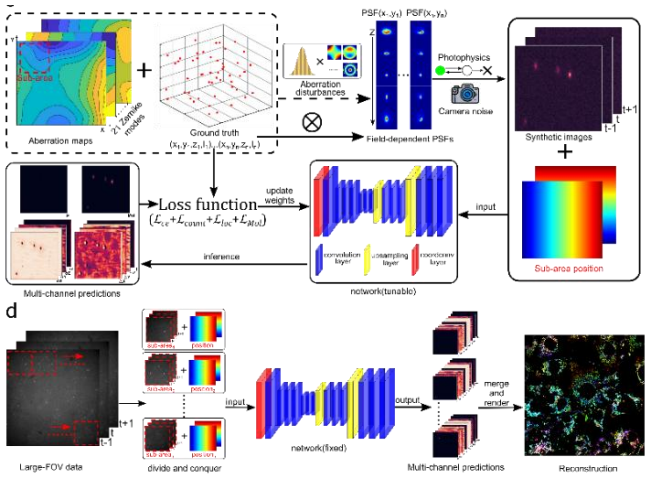
Yiming Li, PhD

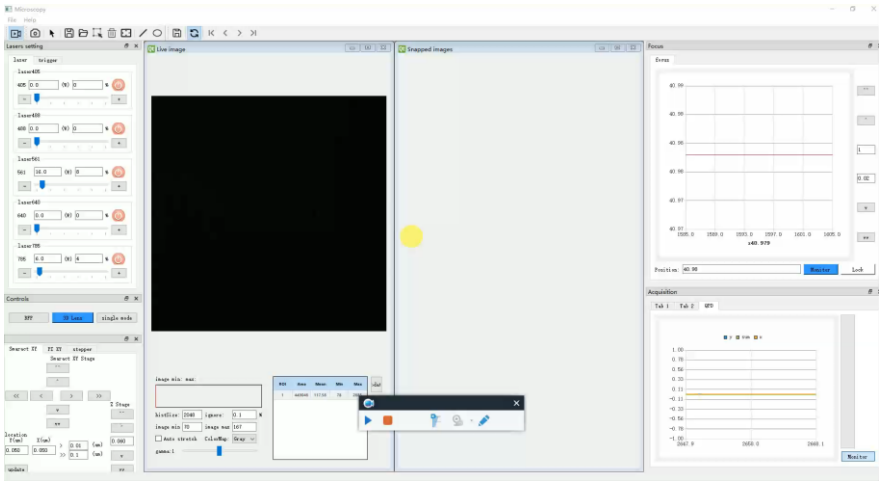
Associate Professor of Biomedical Engineering

Software Repositories

https://github.com/Li-Lab-SUSTech





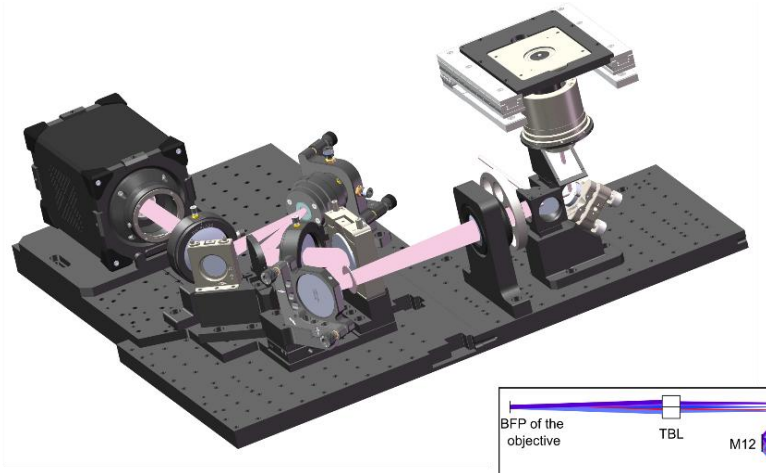


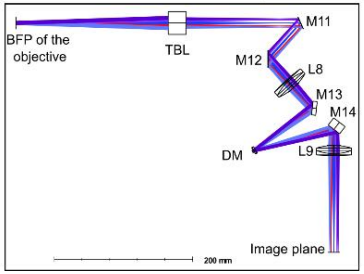


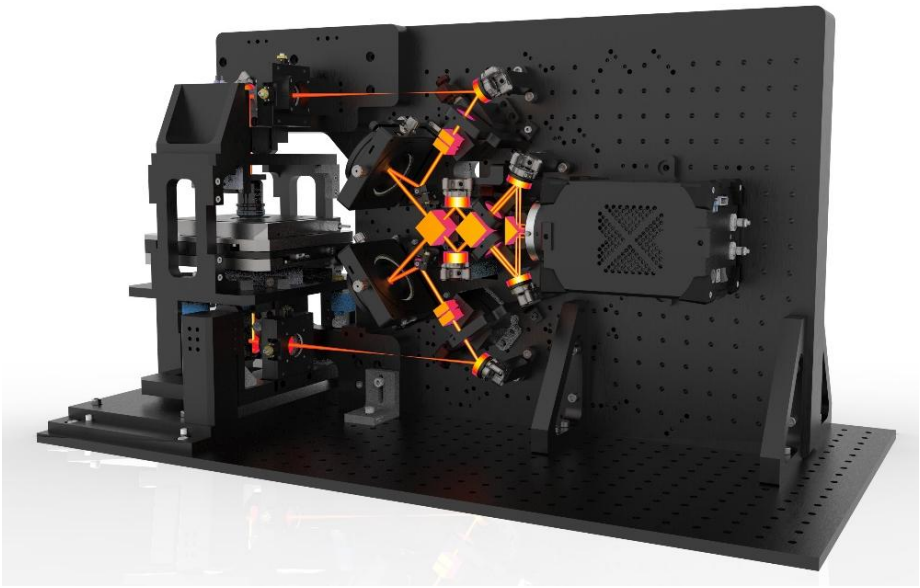
Yiming Li, PhD

Associate Professor of Biomedical Engineering

Microscope Hardware







Self Introduction



群聊：2026医学图像处理

该二维码7天内(9月18日前)有效，重新进入将更新

- Where are you from?
- Which coding languages do you use?
- How many lines of code have you written?
- What do you want to learn from this lecture?

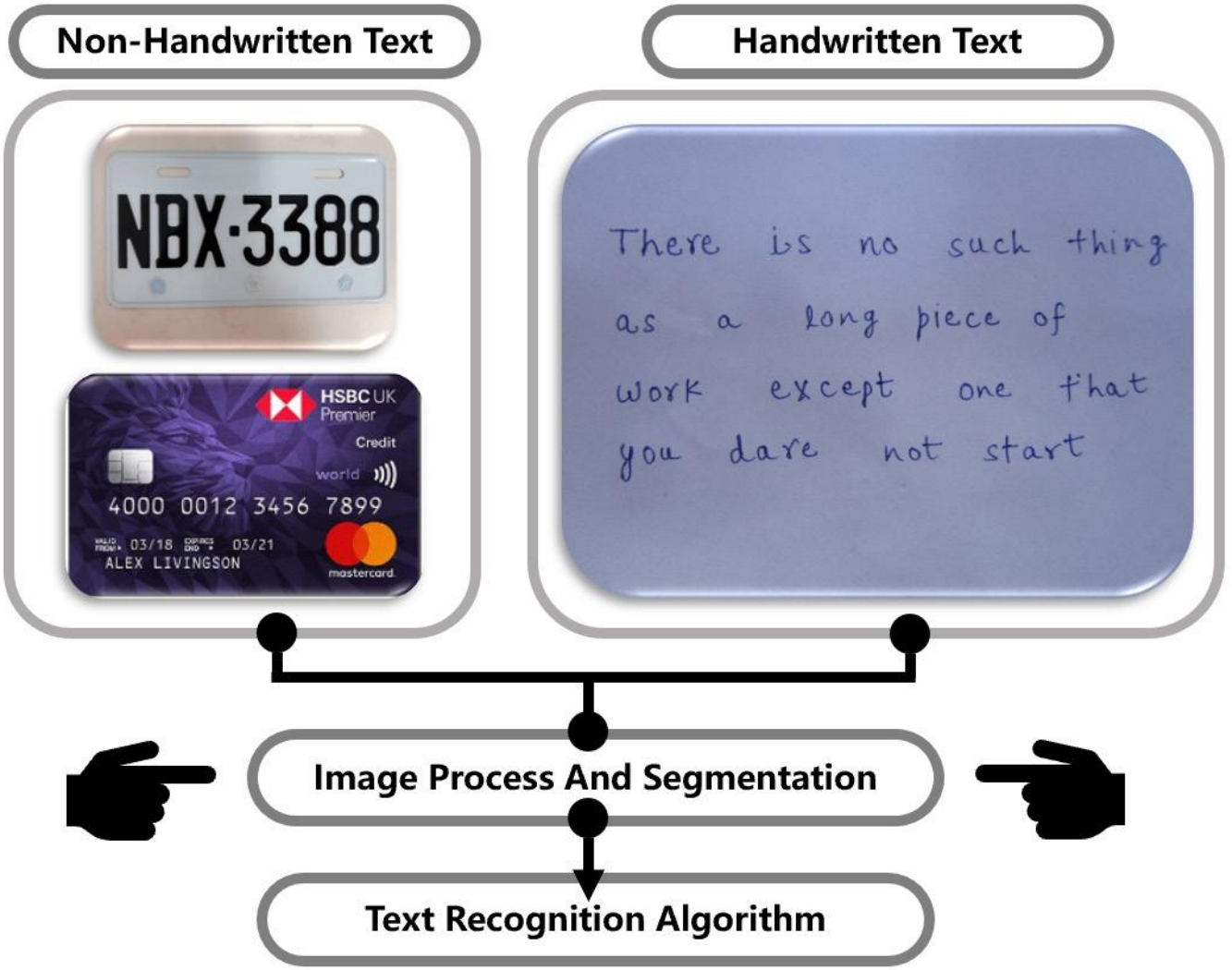
Assessment

- Attendance: 5%
- Quiz: 10%
- Projects : 30% (Week 1-16)
- Final Exam: 40%
- Final Presentation: 15%

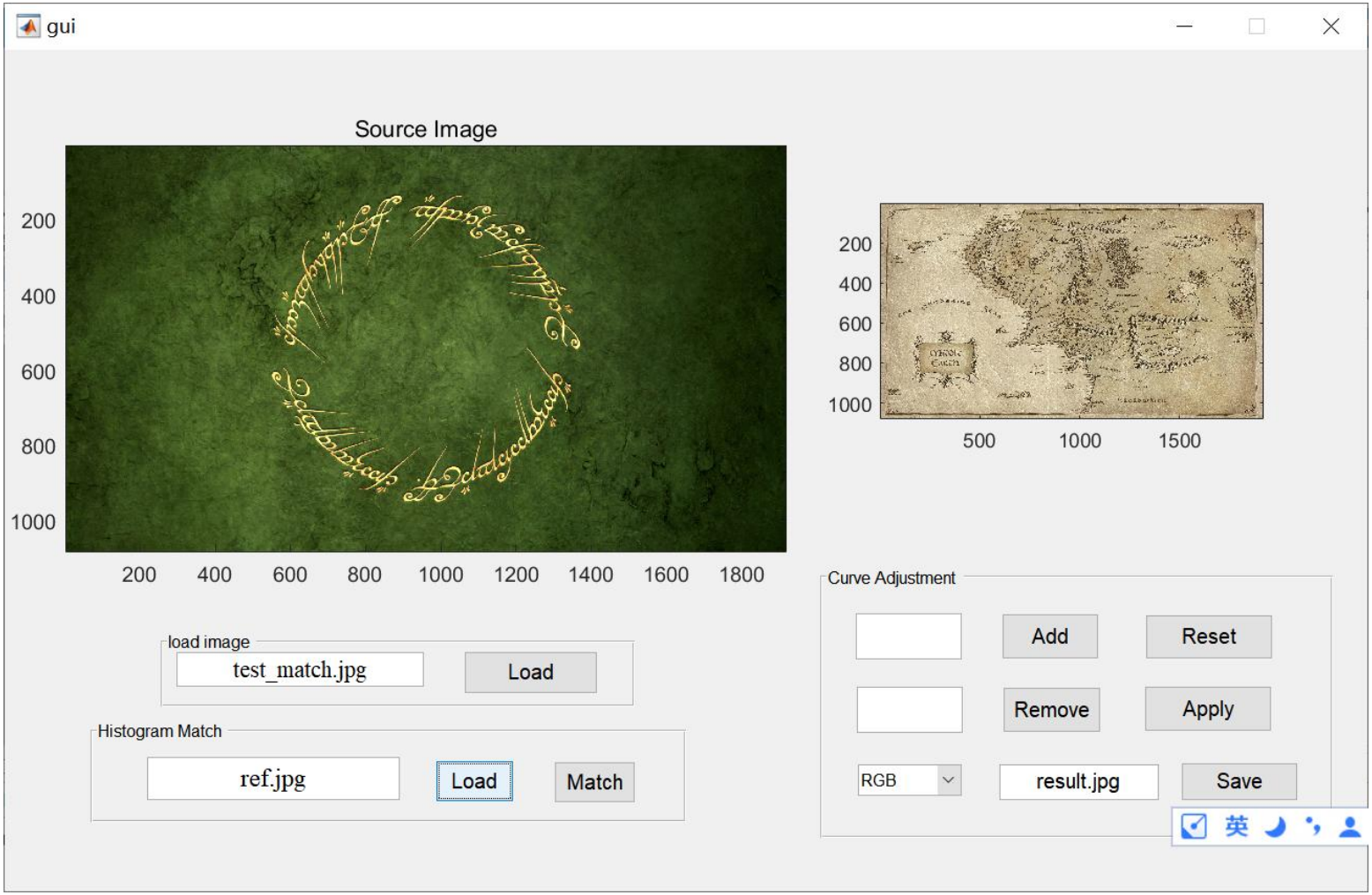
Final Project Outline

- Introduction
- Description of the functionality of the software
- Discussion of the image processing techniques involved
- Technical implementation details
- Summary

Projects Examples

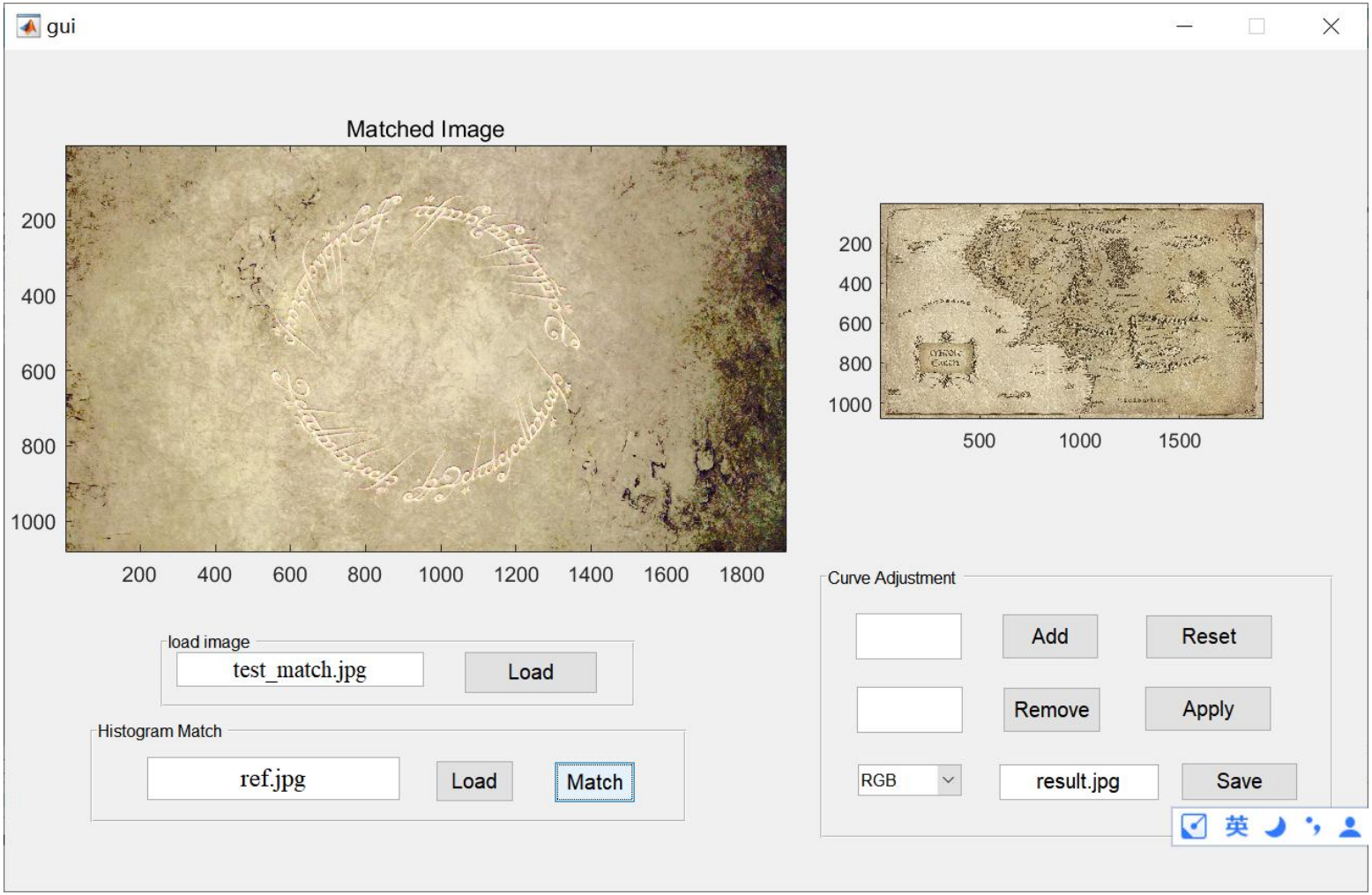


Projects Examples



Extract color themes

Projects Examples



Extract color themes

Projects Examples



9月	暑 假	31 十九	1 二十	2 廿一	3 廿二	4 廿三	5 廿四	6 廿五	9月25日 中秋节	9月3日 开学典礼 9月10日 全校教师大会暨教师表彰大会 9月23日 本科招生工作会议	9月4日 在校本科生返校 9月7日 本科生上课 9月7日-24日 本科生秋季学期退补选课暨导师指导周（1-3周） 9月20日 上单周周五的课	9月4日 在校研究生返校 9月7日 研究生上课 9月20日 上单周周五的课
	第1周 秋季学期	7 白露	8 廿七	9 廿八	10 教师节	11 八月	12 初二	13 初三				
	第2周 秋季学期	14 初四	15 初五	16 初六	17 初七	18 初八	19 初九	20 初十				
	第3周 秋季学期	21 十一	22 十二	23 秋分	24 十四	25 中秋节	26 十六	27 十七				
10月	第4周 秋季学期	28 十八	29 十九	30 二十	1 国庆节	2 廿二	3 廿三	4 廿四	10月1日 国庆节	10月1日 国庆升旗仪式	10月10日 上单周周三的课 10月26日-11月8日 本科生期中考试周（不停课）	10月10日 上单周周三的课 10月26日-11月8日 研究生期中考试周（不停课） 10月-12月 2027级博士生招生录取（第一批）
	第5周 秋季学期	5 廿五	6 廿六	7 廿七	8 寒露	9 廿九	10 九月	11 初二				
	第6周 秋季学期	12 初三	13 初四	14 初五	15 初六	16 初七	17 初八	18 重阳节				
	第7周 秋季学期	19 初十	20 十一	21 十二	22 十三	23 霜降	24 十五	25 十六				
	第8周 期中考试周	26 十七	27 十八	28 十九	29 二十	30 廿一	31 廿二	1 廿三				
11月	第9周 期中考试周	2 廿四	3 廿五	4 廿六	5 廿七	6 廿八	7 立冬	8 三十		11月3日 学术委员会第二次长聘与晋升评估会议 11月8日-11月10日 2026年国际咨询顾问委员会会议 11月13日 校学位评定委员会第四次会议 11月20日-21日 第十二届秋季运动会	11月20日 停课一天	11月20日 停课一天
	第10周 秋季学期	9 十月	10 初二	11 初三	12 初四	13 初五	14 初六	15 初七				
	第11周 秋季学期	16 初八	17 初九	18 初十	19 十一	20 十二	21 十三	22 小雪				
	第12周 秋季学期	23 十五	24 十六	25 十七	26 十八	27 十九	28 二十	29 廿一				
12月	第13周 秋季学期	30 廿二	1 廿三	2 廿四	3 廿五	4 廿六	5 廿七	6 廿八		12月4日 第六届校园公益文化周 12月5日 科研工作会议 12月17日 校学位评定委员会第五次会议 12月26日 校友会理事会会议、第八届校友年会	12月28日-1月8日 本科生期末复习考试周	12月28日-1月8日 研究生期末复习考试周 12月下旬 全国硕士研究生招生考试
	第14周 秋季学期	7 大雪	8 三十	9 冬月	10 初二	11 初三	12 初四	13 初五				
	第15周 秋季学期	14 初六	15 初七	16 初八	17 初九	18 初十	19 十一	20 十二				
	第16周 秋季学期	21 十三	22 冬至	23 十五	24 十六	25 十七	26 十八	27 十九				
	第17周 复习考试周	28 二十	29 廿一	30 廿二	31 廿三	1 元旦	2 廿五	3 廿六				

Textbook and Supplementary Readings

教材： Very popular in the world!

- R.C.Gonzalez, R.E. Woods, Digital Image Processing, 第三版，电子工业出版社，2010年

http://www.imageprocessingplace.com/



PhD: 1970
IEEE Fellow

Books:
1977, 1992
2002, 2008
2010



主要参考教材:

1. **R.C.Gonzalez, R.E. Woods, Digital Image Processing using MATLAB** 电子工业出版社, 2005年
2. 章毓晋，图像工程(上册)：图像处理，清华大学出版社，2006年



What Is MATLAB?

Getting Started

What Is MATLAB?

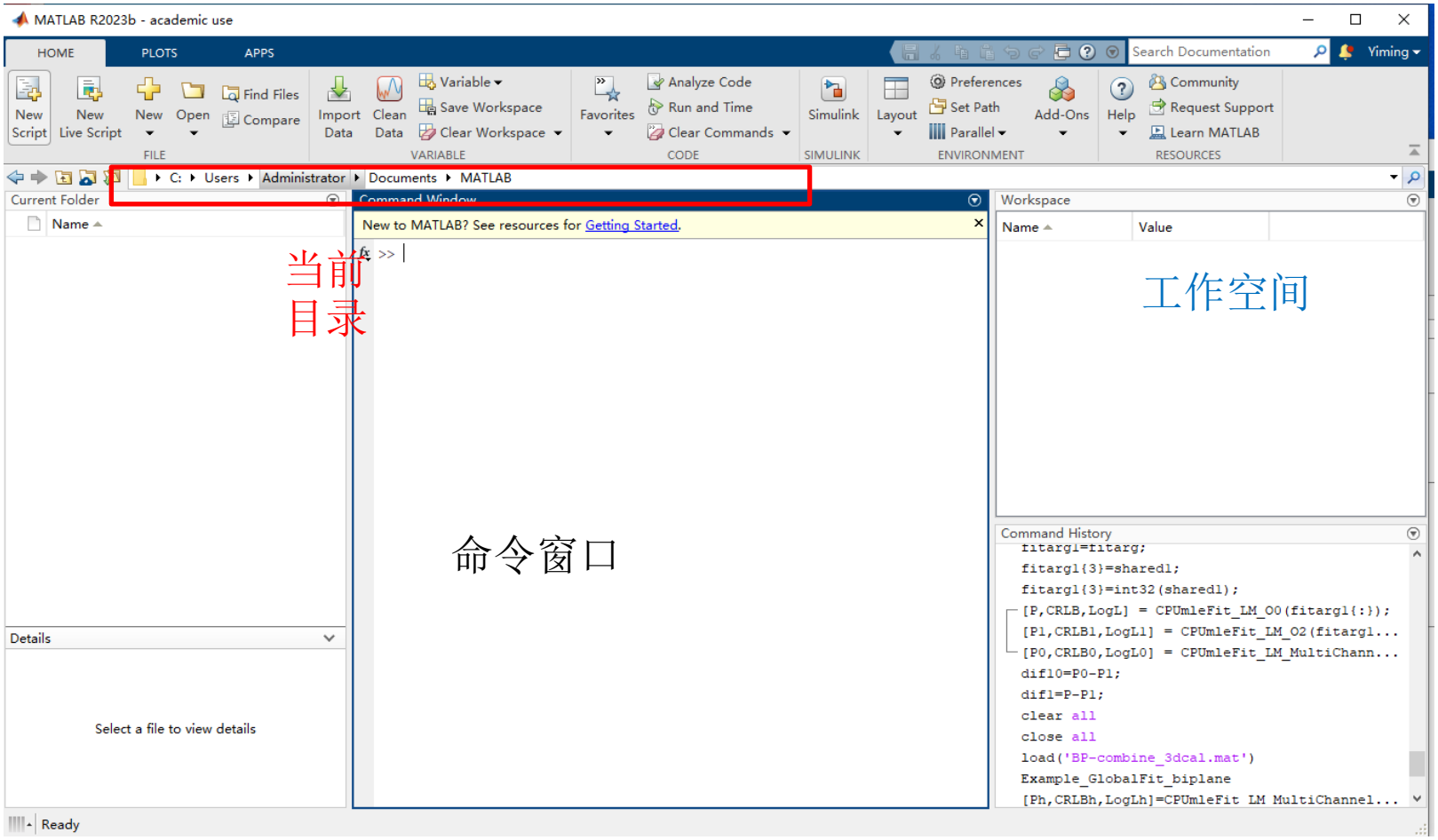
MATLAB is a high-performance language for technical computing. It integrates computation, visualization, and programming in an easy-to-use environment where problems and solutions are expressed in familiar mathematical notation. Typical uses include

- Math and computation
- Algorithm development
- Data acquisition
- Modeling, simulation, and prototyping
- Data analysis, exploration, and visualization
- Scientific and engineering graphics
- Application development, including graphical user interface building

MATLAB is an interactive system whose basic data element is an array that does not require dimensioning. This allows you to solve many technical computing problems, especially those with matrix and vector formulations, in a fraction of the time it would take to write a program in a scalar noninteractive language such as C or Fortran.

The name MATLAB stands for *matrix laboratory*. MATLAB was originally written to provide easy access to matrix software developed by the LINPACK and EISPACK projects. Today, MATLAB engines incorporate the LAPACK and BLAS libraries, embedding the state of the art in software for matrix computation.

What Is MATLAB？





Quiz 1

- What is the command to read images in Matlab?

A. imtool

B. imageread

C. image

D. imread

Quiz 2

- What is the command to change the image size in Matlab?

A. imtool

B. imresize

C. resize

D. imread

Quiz 3

- What is the first word of a function in Matlab script?

A. function
B. imlist
C. image
D. imread



Quiz 4

- What does * mean in Matlab matrix operations?

A. Multiplying matrix elements

B. Matrix multiplication

Image Analysis Software




- ImageJ - Fiji



- Matlab



- Ilastik



- Cell profiler





- Languages (Python, R, Java…)

百闻不如一见

One picture is worth more than ten thousand words.

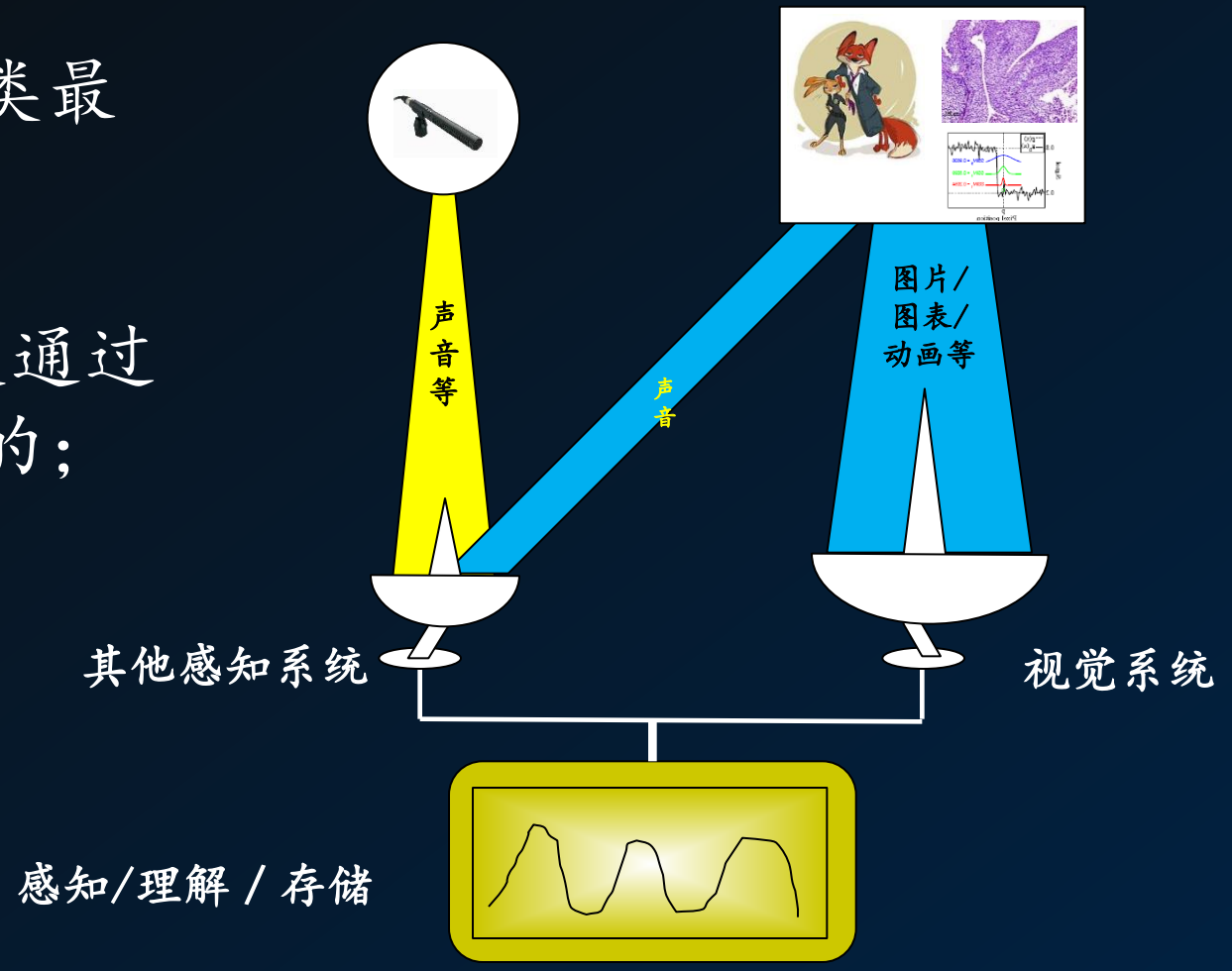
Anonymous

《汉书·赵充国传》：“百闻不如一见，兵难遥度，臣愿驰至金城，图上方略。”

百闻不如一见。对方军事上的情况如何，在后方很难准确地估计，还是让我上前方金城了解了解再来制定策略吧！

视觉系统是人类最高级的感知；

约75%的信息是通过视觉系统获取的；



白金还是蓝黑？



Hollywood Life

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What Color Is This Dress? Taylor Swift, Lucy Hale & More Celebs React

Fri, February 27, 2015 12:18am EST by jordynshaffer

7 Comments 15,266 Article Views

Share 2.5k Tweet 55 +1 1



The INDEPENDENT



The dress can be seen in different colours

White and gold or blue and black? The dress has confused the internet but science has the answer

The Washington Post

The inside story of the 'white dress, blue dress' drama that divided a planet

By Terrence McCoy February 27 at 1:57 AM

Follow @terrence_mccoy



DAILY NEWS

U.S.

'The dress' has everyone arguing over its colors

ARIELLE LANDAU, JASON SILVERSTEIN

Yesterday, 9:21 PM

swiked
officialunitedstates
tumblr. Follow



TIME

The Dress That Broke the Internet, and the Woman Who Started It All

Rishi Iyengar @iyengarrishi

4:07 AM ET

SHARE

TAP



Caitlin ... start s... by Taylor ...

MailOnline News

What colors are this dress? White and gold or blue and black? Fierce internet debate is sparked by two-tone outfit - and even Kim and Kanye are divided!

By Chris Spargo For Dailymail.com and Stephanie Linning for MailOnline

01:50 27 Feb 2015, updated 10:19 27 Feb 2015



白金还是蓝黑？



Introduction

1. 什么是数字图像处理

   What is digital image processing?

2. 数字图像处理的起源

   The origins of digital image

3. 使用数字图像处理领域的实例

   Examples of fields that use digital image processing

4. 数字图像处理的基本步骤

   Fundamental steps in digital image processing

5. 数字图像处理系统组成

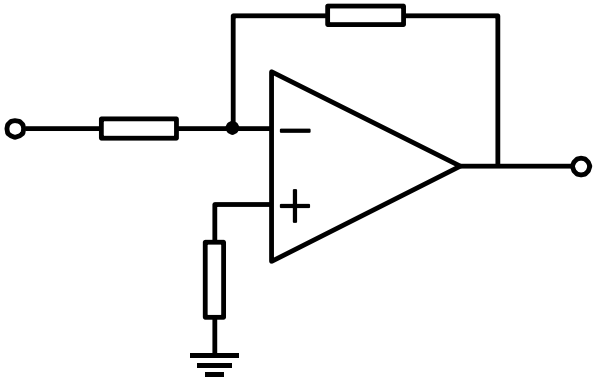
   Components of an image processing system

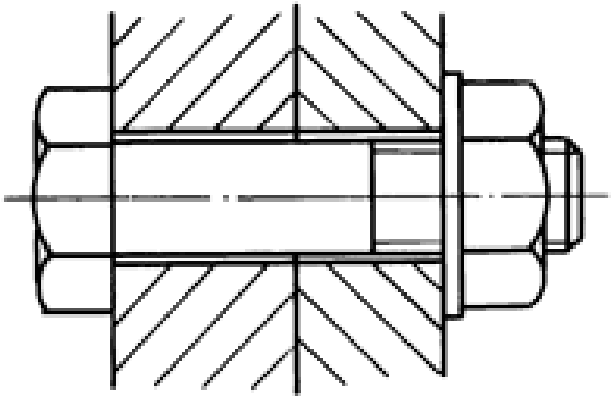
Pictures





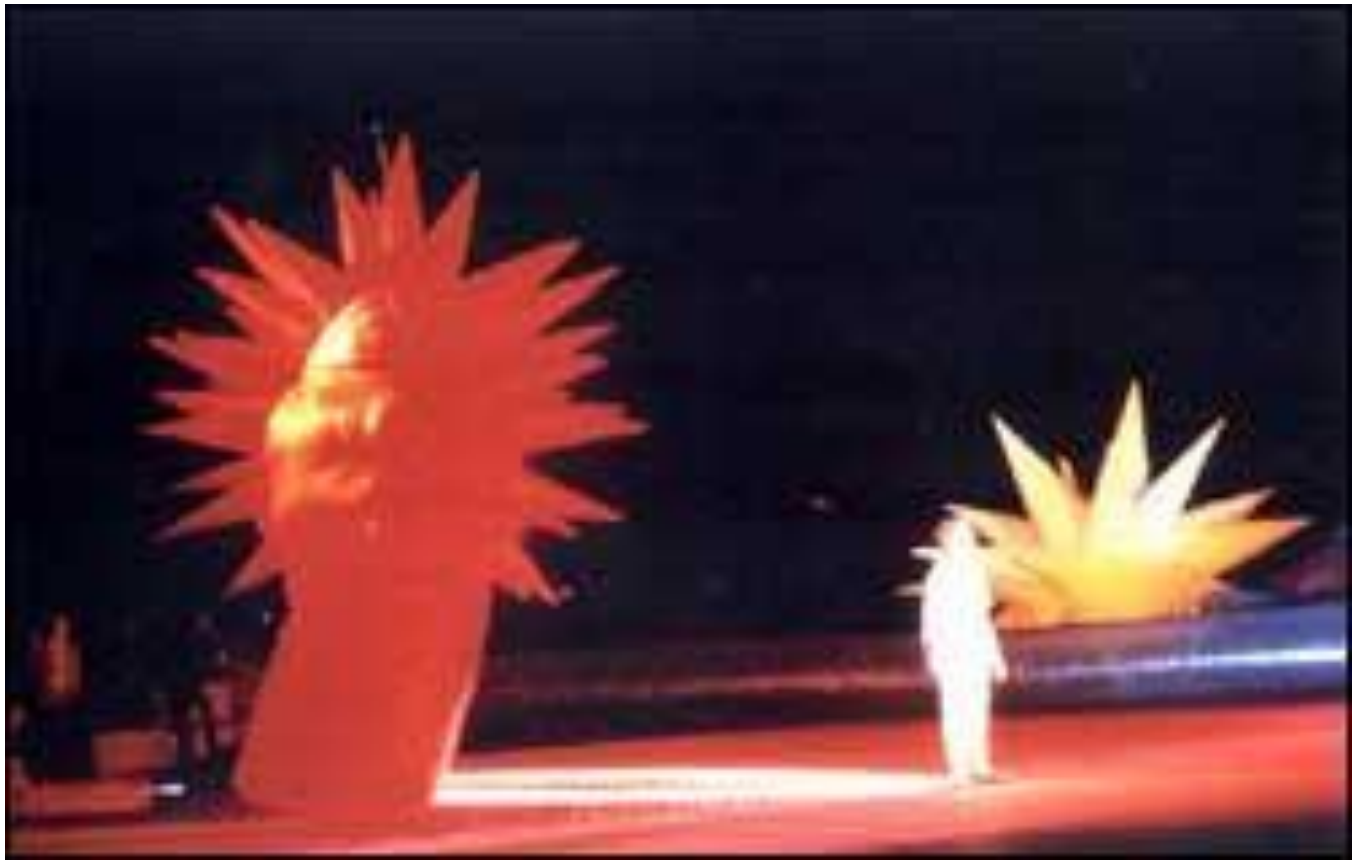
1	2
3	4



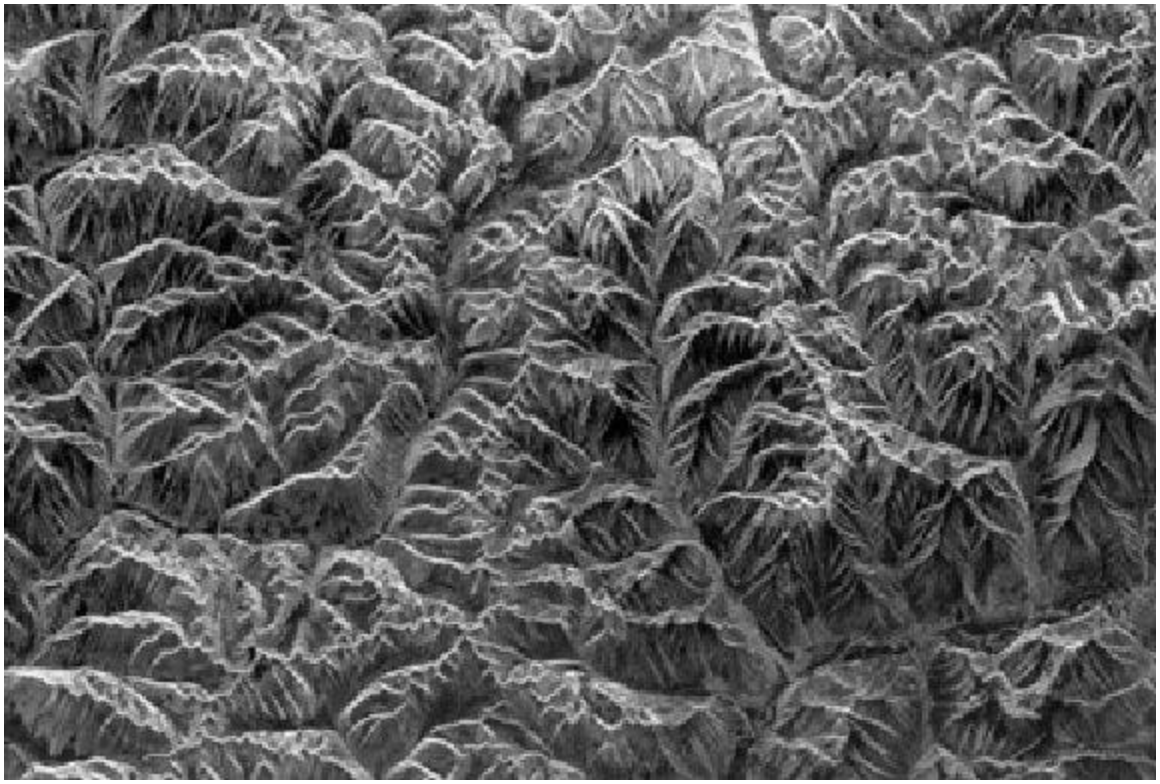


1. Painting
2. Photo
3,4. Drawings

Optical Image



Non-Visible Images



Spaceborne radar image of mountains in southeast Tibet

Image of an Object

- Defined as a two-dimensional function *f(x,y),* where $x, y \in R$ are space (plane) coordinates, and the amplitude $f \in R$ at the coordinates (x, y) called the **gray level** or **intensity** of the image.
- **Gray level image:** $f(x, y)$ represents gray level information
- **Color image**: $f(x, y, channel)$ is a three-dimensional function, where channel represents the color model used by the color image, and RGB is commonly used.

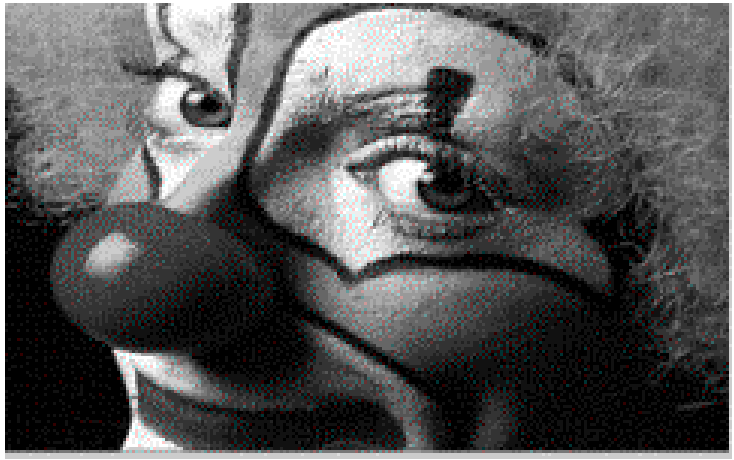
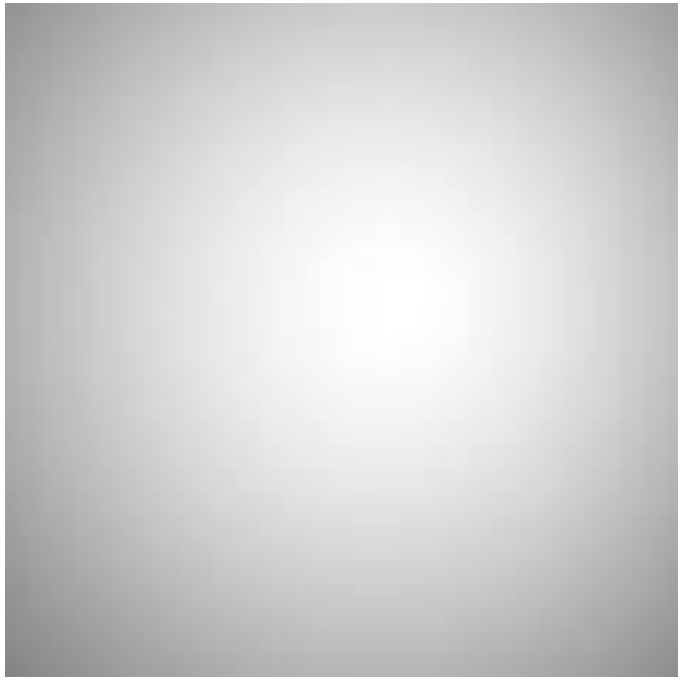
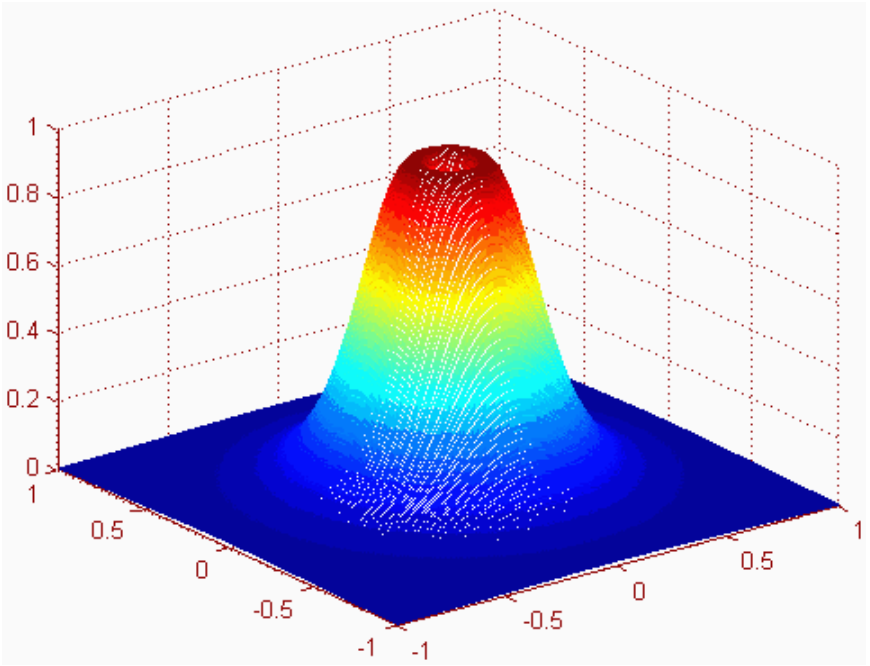




Image of an Object

Mathematical Functions - Continuous





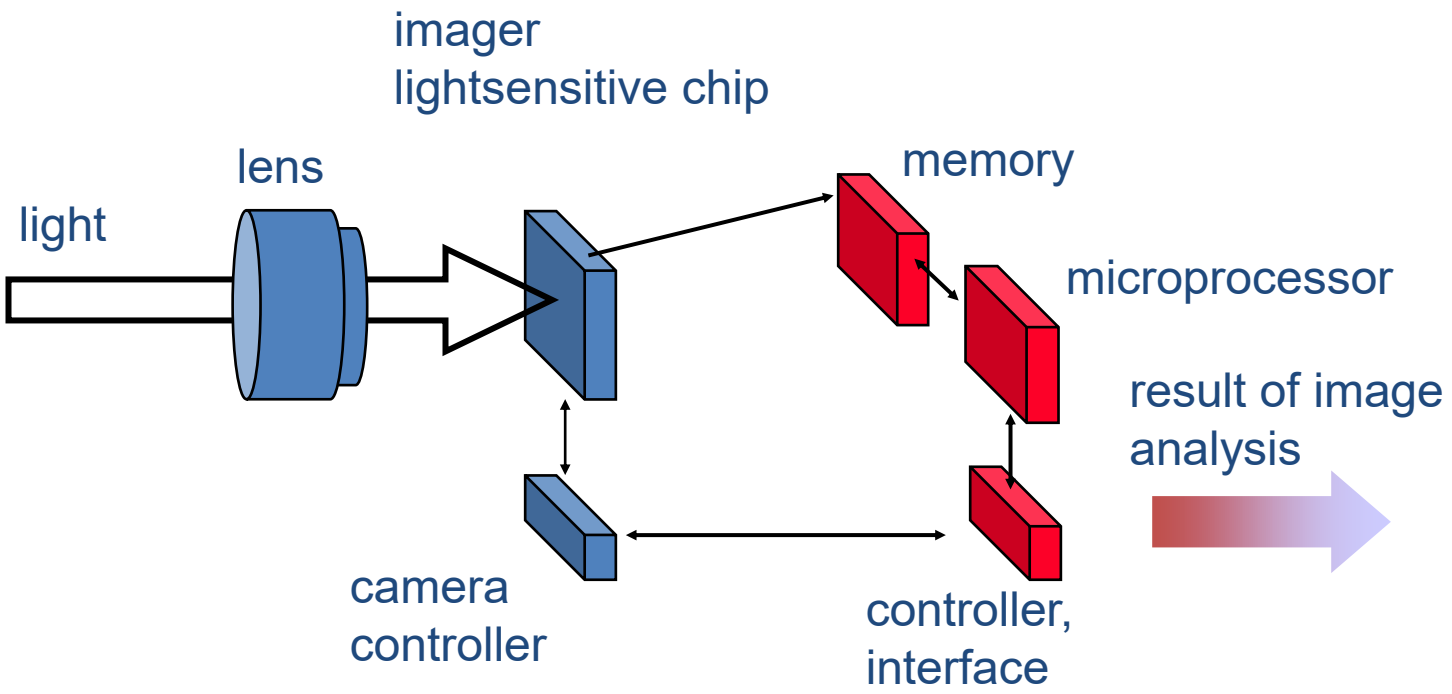
$$I_F = \cos\varphi \exp\left[-\mu_A D_A(x,y) - \mu_F T_F / \cos\varphi\right]$$

$$H(u,v) = \frac{1}{1+\left[\frac{D^2(u,v)-D_0^2}{D(u,v)W}\right]^{2n}}$$

Image Formation and Analysis

electronic camera (formation)

ECU (electronic control unit) (image processing)



Imager

- Process of image formation:
  - incident light intensity:

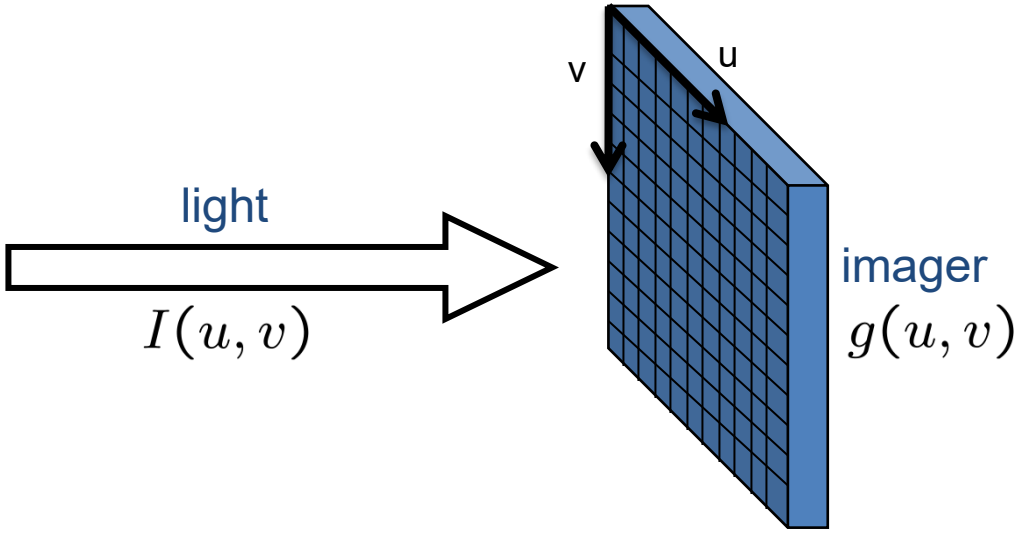
    $I : \mathbb{R}^2 \rightarrow \mathbb{R}$

  - output of imager:

$g : \{0, \ldots, w-1\} \times \{0, \ldots, h-1\} \rightarrow \{0, \ldots, g_{max}\}$

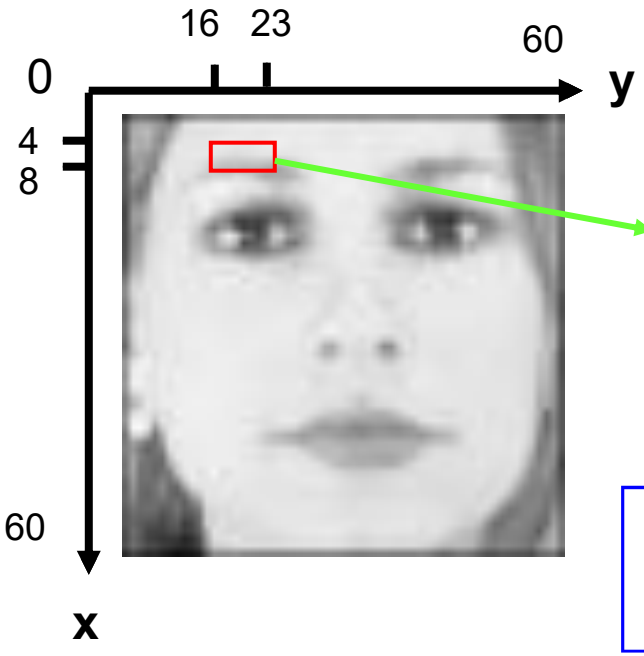
w, h : image width, height

Features: Both the light position and light intensity change continuously



Digital Image

- The image's spatial **coordinate** information and **gray level (intensity)** are both **discrete** and **quantized** to obtain an image that can be processed by a computer
- A digital image is composed of a **limited** number of elements, each of which has a **specific position** (spatial coordinates) and **amplitude**.



232	232	232	233	234	235	236	236
230	230	231	233	234	234	233	234
221	223	226	228	229	227	226	228
197	197	196	195	198	195	195	205
187	181	178	175	175	168	170	180

✓Pixel(像素): Picture Element

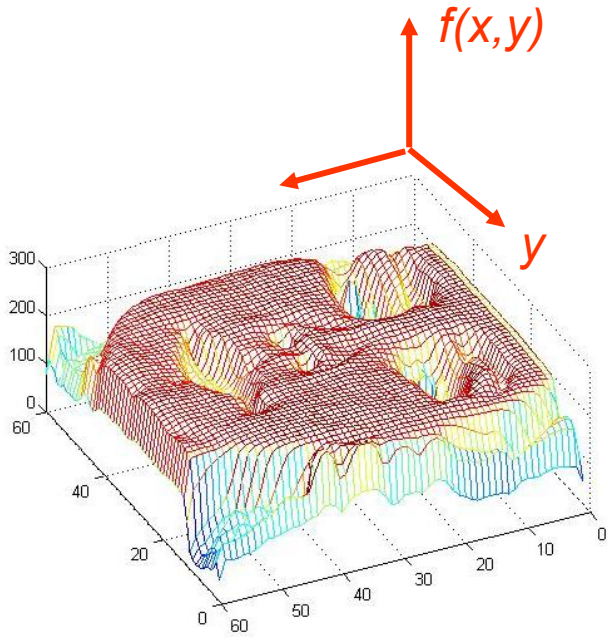
✓$f(x, y)$ Represent brightness (gray level) at one point

Digital Image

Mathematical Functions - Discrete

- **Digital image: an image with coordinates and gray level discretization**
- It is defined as a two-dimensional function $f(x, y)$, where $x \in \mathrm{M}$ and $y \in \mathrm{N}$ represent space (plane) coordinates, and the amplitude $f \in \mathrm{N}$ at coordinates (x, y) is called gray level or intensity.





$$I = f(x,y) = \begin{bmatrix} i_{0,0} & i_{0,1} & \cdots & i_{0,N-1} \\ \vdots & \vdots & \vdots & \vdots \\ i_{M-1,0} & i_{M-1,1} & \cdots & i_{M-1,N-1} \end{bmatrix}_{M*N}$$

What is digital image processing?

- Please list the digital images in life.
- Computer display screen, printer, scanner, mobile phone, photo, jpg or avi on the computer, two-dimensional code. . .

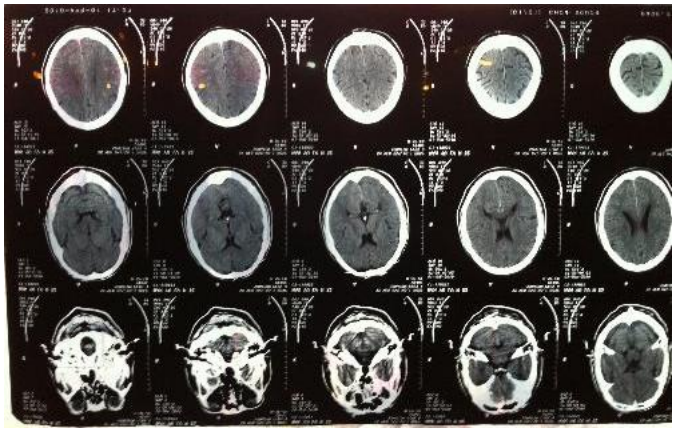
- Imagine what the world would look like if these were used in analog form?

Biomedical image processing

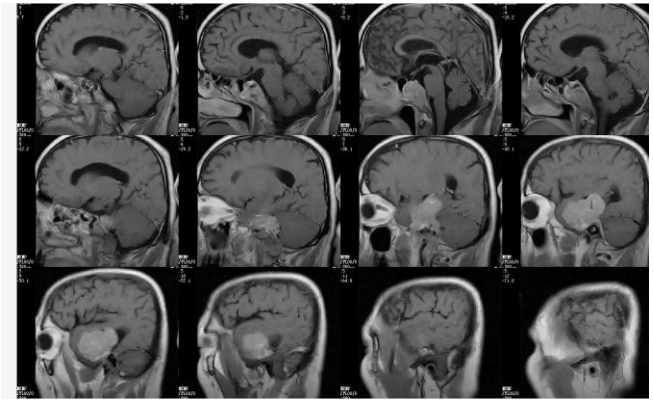
- Please list the digital images of the hospital



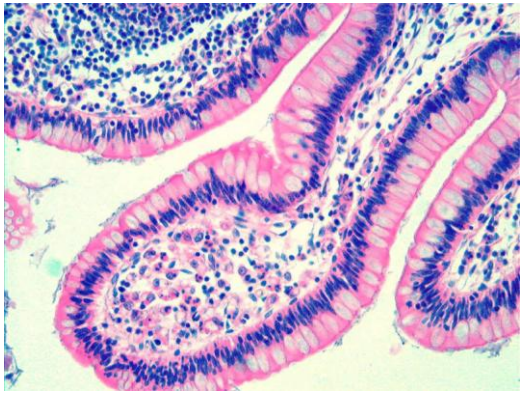
X光片



CT



MRI



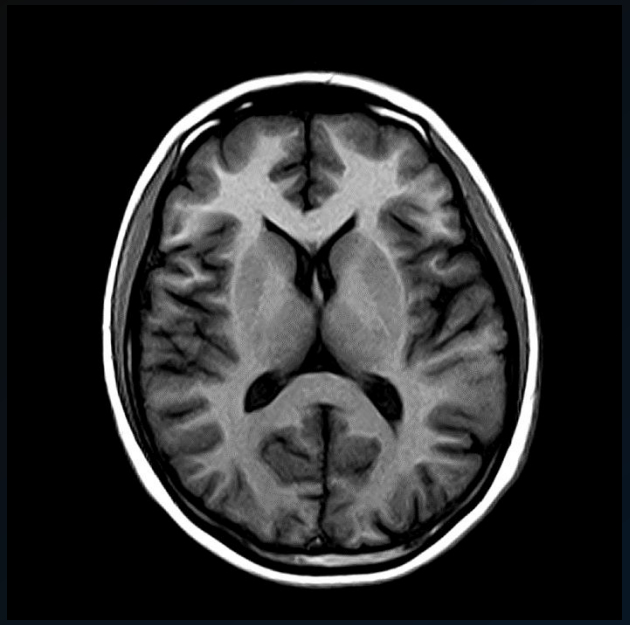
H&E染色



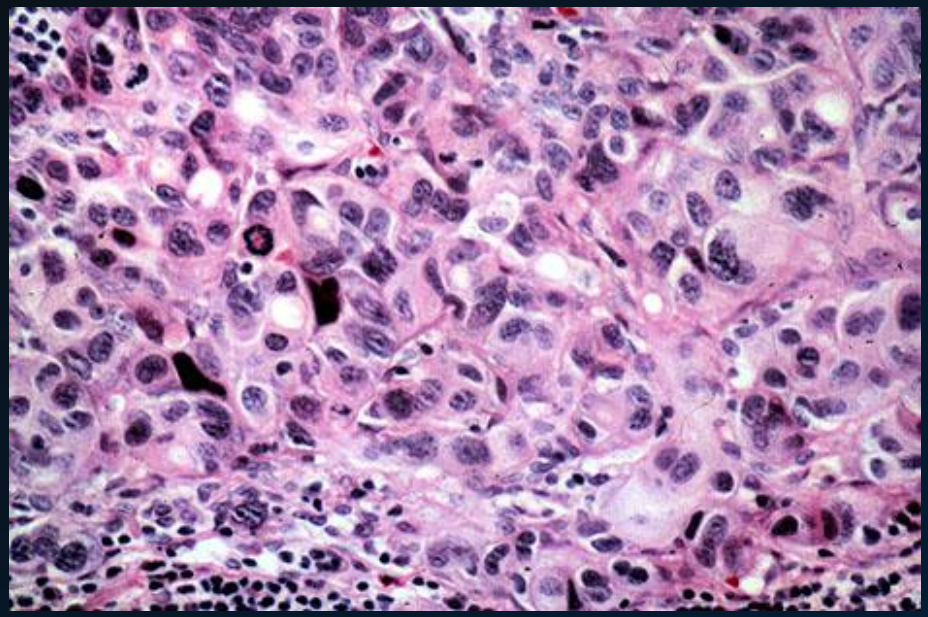








MRI



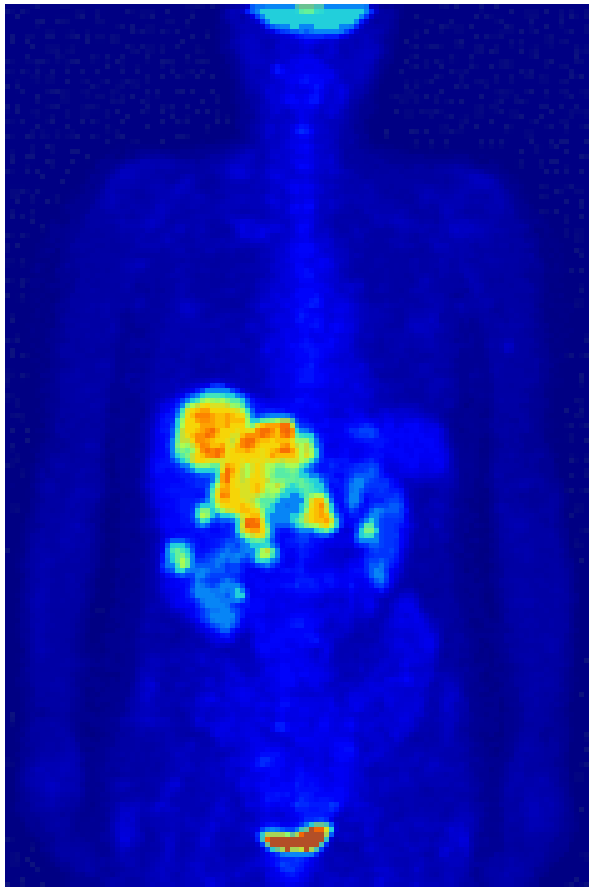
病理切片



眼底



内窥镜



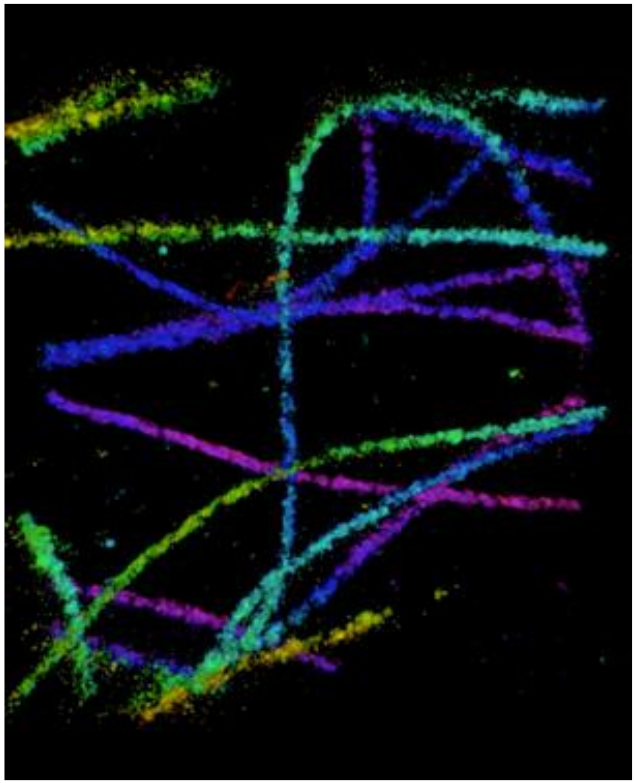
PET



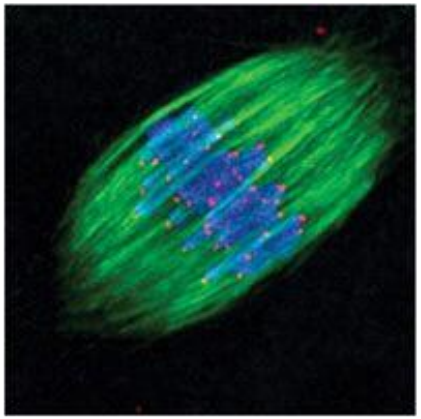
超声



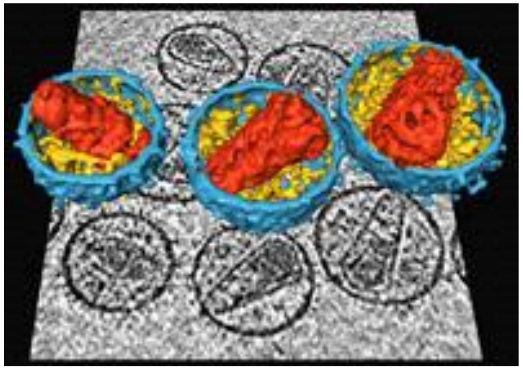
Image Analysis in Science



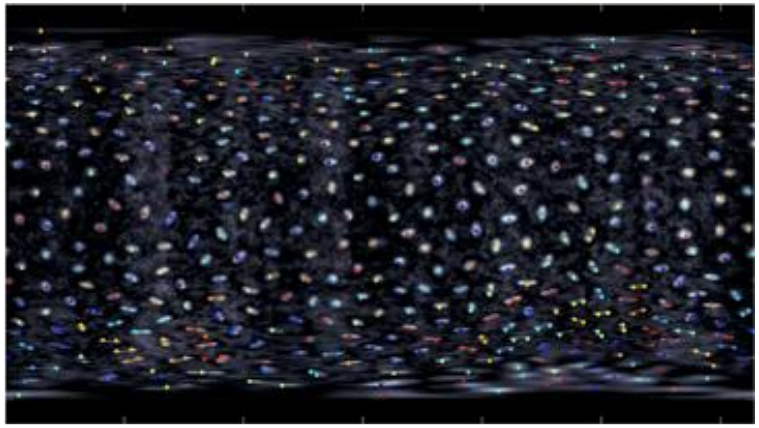
Microtubules



Meiotic spindle



3D reconstruction of HIV-1



Drosophila embryo

What is digital image processing?

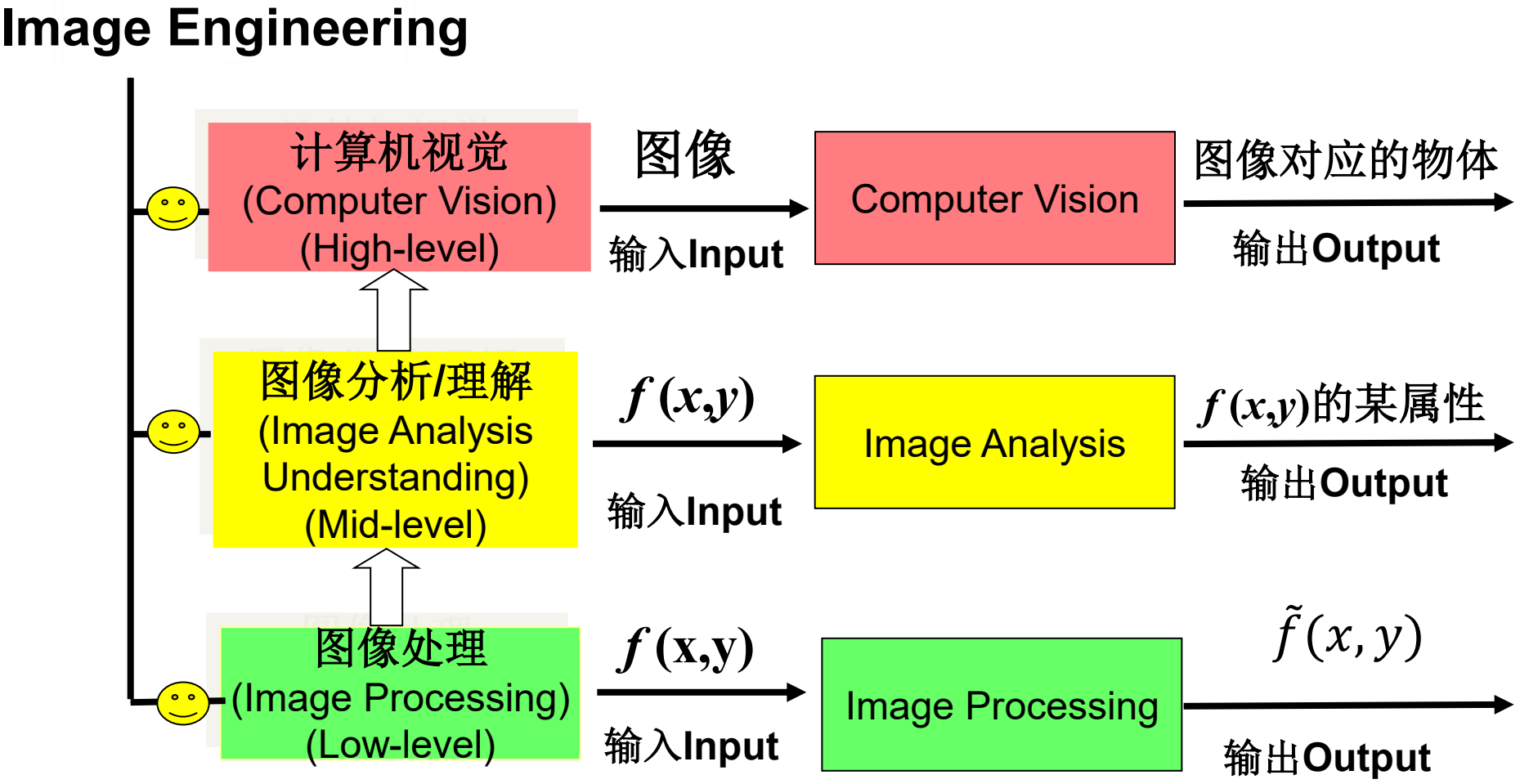
Image science is a subject with great depth
Image processing is only one level of image science





What is digital image processing?

Three levels of image engineering



Three levels of image engineering

- **Simple point calculation (low-level processing)**



Image→Image

Three levels of image engineering

- **Denoise (low-level processing)**



Degraded Image



Noise reduced Image

Image→Image

Three levels of image engineering

- **Image Enhancement (low-level processing)**



Observed Image

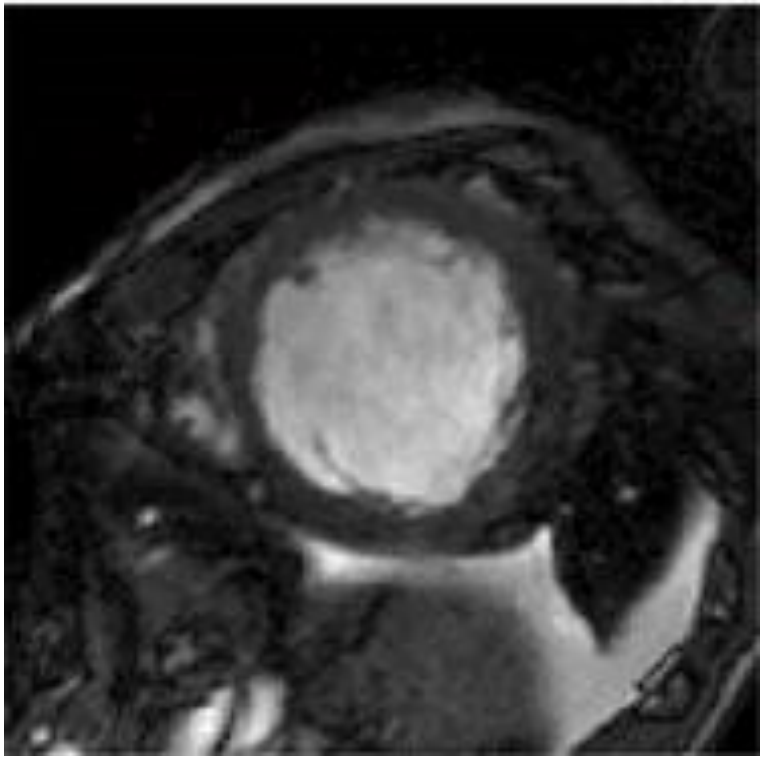


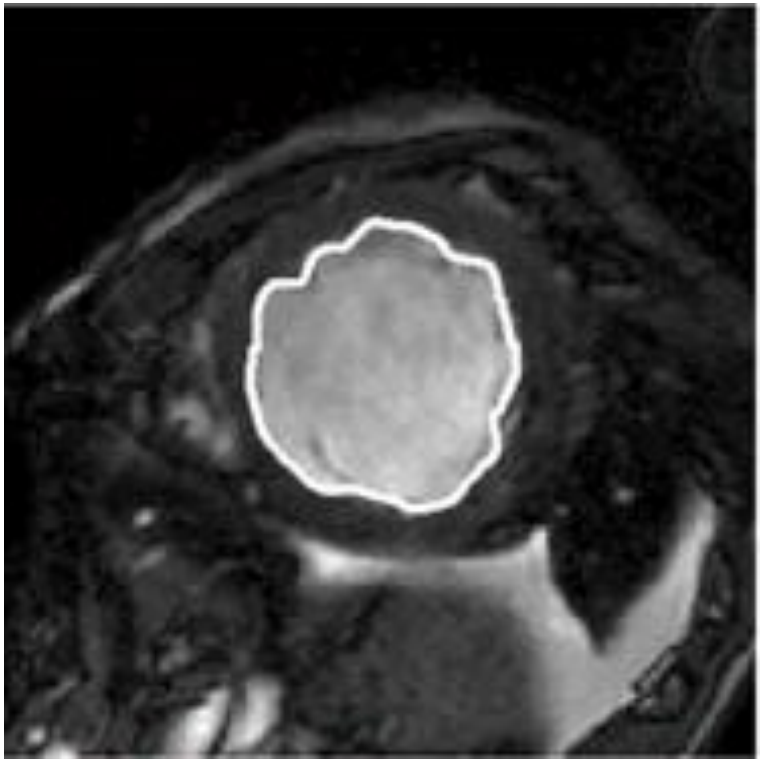
Enhanced Image

Image→Image

Three levels of image engineering

- **Image Segmentation (mid-level processing)**





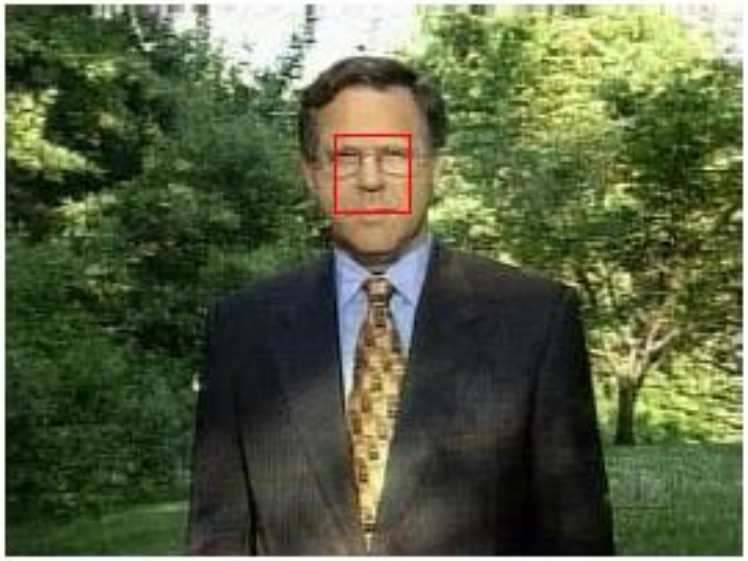
The size of the heart chamber volume

Image→data

Three levels of image engineering

- **Face tracking and recognition (high-level processing)**





Three Levels of Image Engineering



Edge, Color, Texture, etc.



Introduction

1. 什么是数字图像处理

   What is digital image processing?

2. **数字图像处理的起源**

   The origins of digital image

3. 使用数字图像处理领域的实例

   Examples of fields that use digital image processing

4. 数字图像处理的基本步骤

   Fundamental steps in digital image processing

5. 数字图像处理系统组成

   Components of an image processing system

Origins of Digital Image Processing



FIGURE 1.1 A digital picture produced in 1921 from a coded tape by a telegraph printer with special typefaces. (McFarlane.) [References in the bibliography at the end of the book are listed in alphabetical order by authors' last names.]



FIGURE 1.2
A digital picture made in 1922 from a tape punched after the signals had crossed the Atlantic twice. (McFarlane.)

Origins of Digital Image Processing

It is processed by a digital computer, the input is an image, and the output is also an image. The calculation process is called digital image processing.

“高性能”计算机
(1960s)
(Storage,Display)

太空计划
(1960s)
(漫步者7号)

放射医学
(late 1960s,
early 1970s)
(CT,1979, Nobel Prize)

Digital Image Processing

图像增强
Image Enhancement

图像恢复
Image Restoration



美国航天器传回的第一张月球照片，由漫步者7号在探测器在撞击月球表面前17分钟于美国东部时间1964年7月31日上午9:09拍摄

From Gonzalez’s book, Fig.1.4

Origins of Digital Image Processing

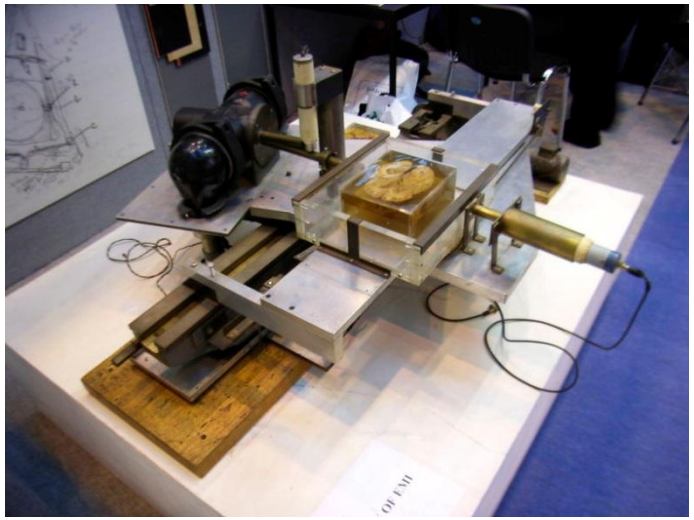
It is processed by a digital computer, the input is an image, and the output is also an image. The calculation process is called digital image processing.

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(1960s)
(Storage,Display)

太空计划
(1960s)
(漫步者7号)

放射医学
(late 1960s,
early 1970s)

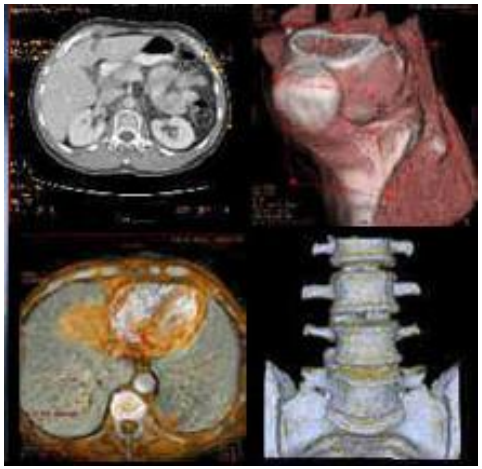
Digital Image Processing



X ray, 1895,
Roentgen. 1901
Nobel Prize in
Physics



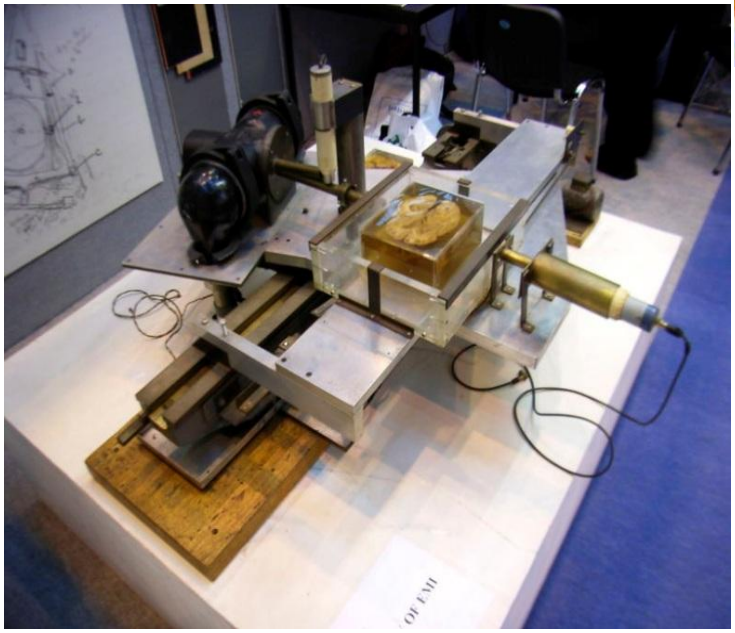
1970, Hounsfield, the first CT
1979, Nobel Prize in Medicine





Origins of Digital Image Processing

It is processed by a digital computer, the input is an image, and the output is also an image. The calculation process is called digital image processing.



1970, Hounsfield, the first CT
1979, Nobel Prize in Medicine



The Nobel Prize in Physiology or Medicine 1979

"for the development of computer assisted tomography"



Allan M. Cormack

1/2 of the prize

1924~1998

Tufts University，
Medford, MA, USA



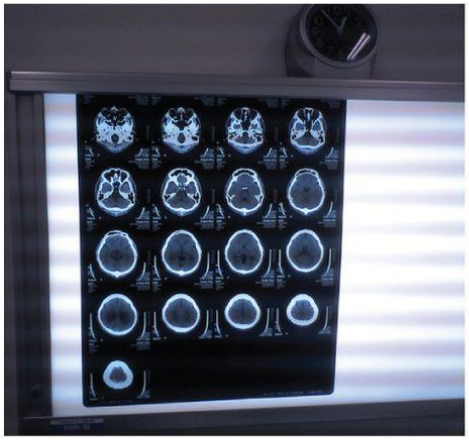
Godfrey N. Hounsfield

1/2 of the prize

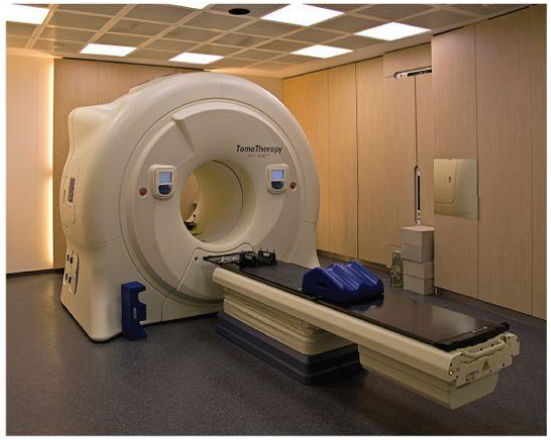
1919~2004

Central Research Laboratories, EMI
London, United Kingdom

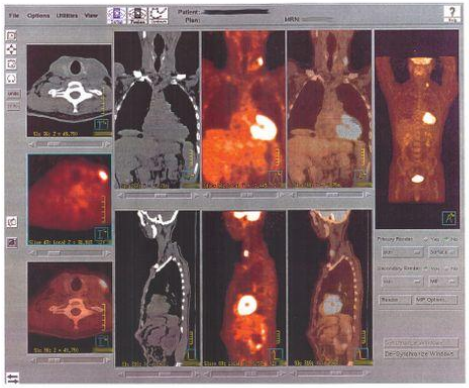
What is Medical Imaging



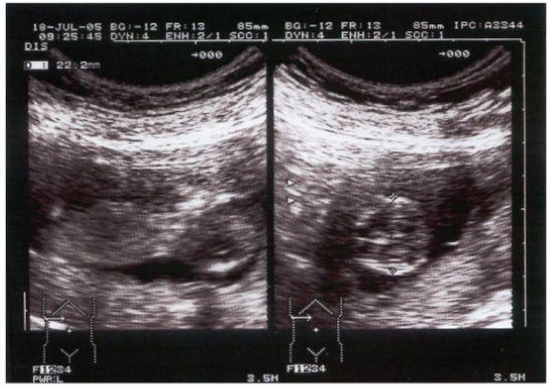
(a)



(b)



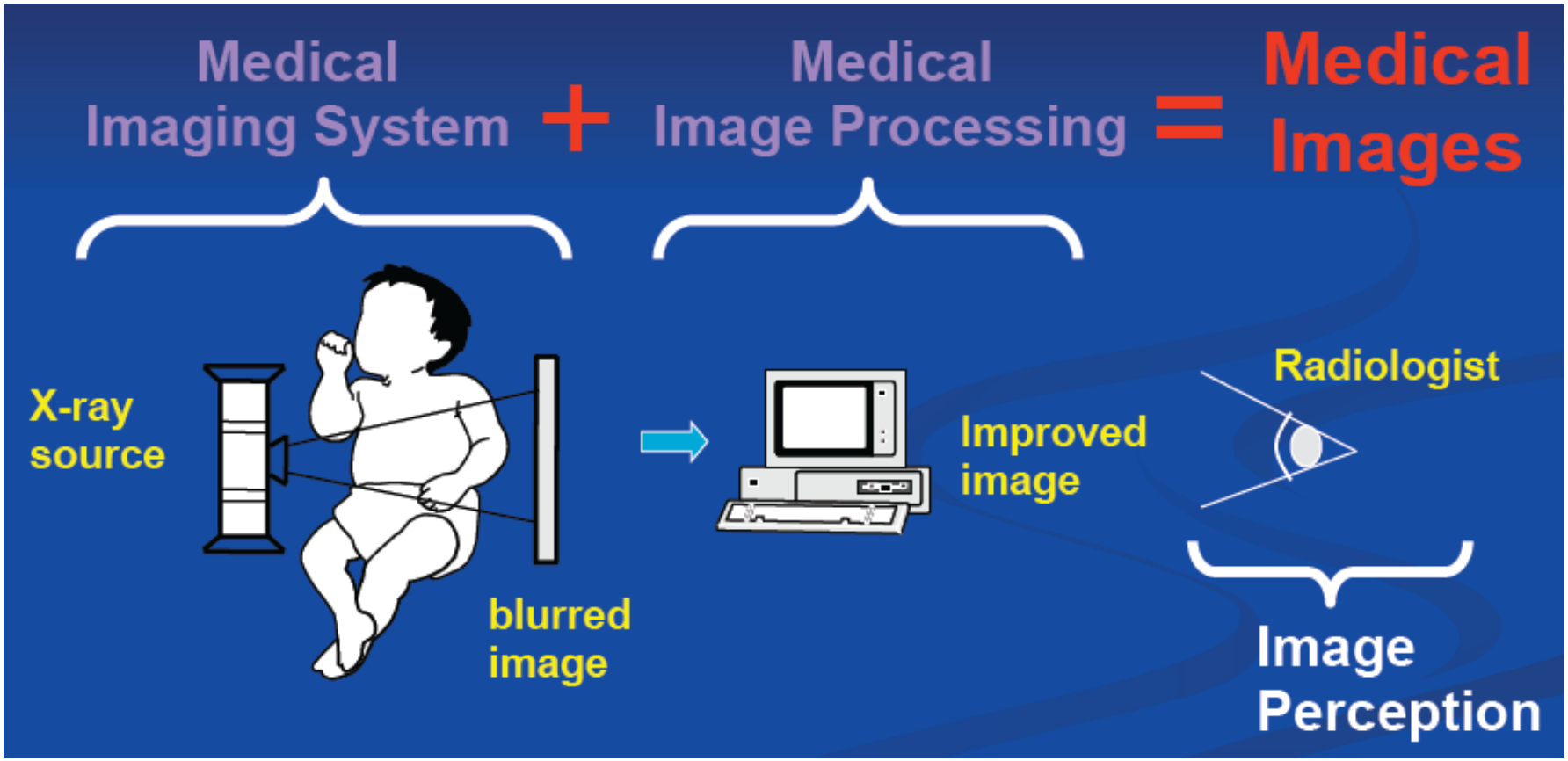
(c)



(d)

- X-ray radiography
- Magnetic resonance imaging (MRI)
- Medical ultrasonography or ultrasound
- Endoscopy
- Elastography
- Tactile imaging
- Thermography
- Medical photography
- Nuclear medicine functional imaging techniques e.g. positron emission tomography (PET)

What is Medical Imaging



Research Area

- 医学成像系统 (Medical Imaging System)
  - 图像形成的物理(化学)原理及其过程
  - 成像机理、成像设备、成像系统分析
- 医学图像处理与分析 (Medical Image Processing and Analysis)
  - 对获得的图像，采用各种算法进一步处理
  - 图像增强、特征提取、分类、分割、识别等
- 医学图像应用研究 (Medical Image Applications)
  - 临床医学应用
  - 基础医学与科学研究

Career

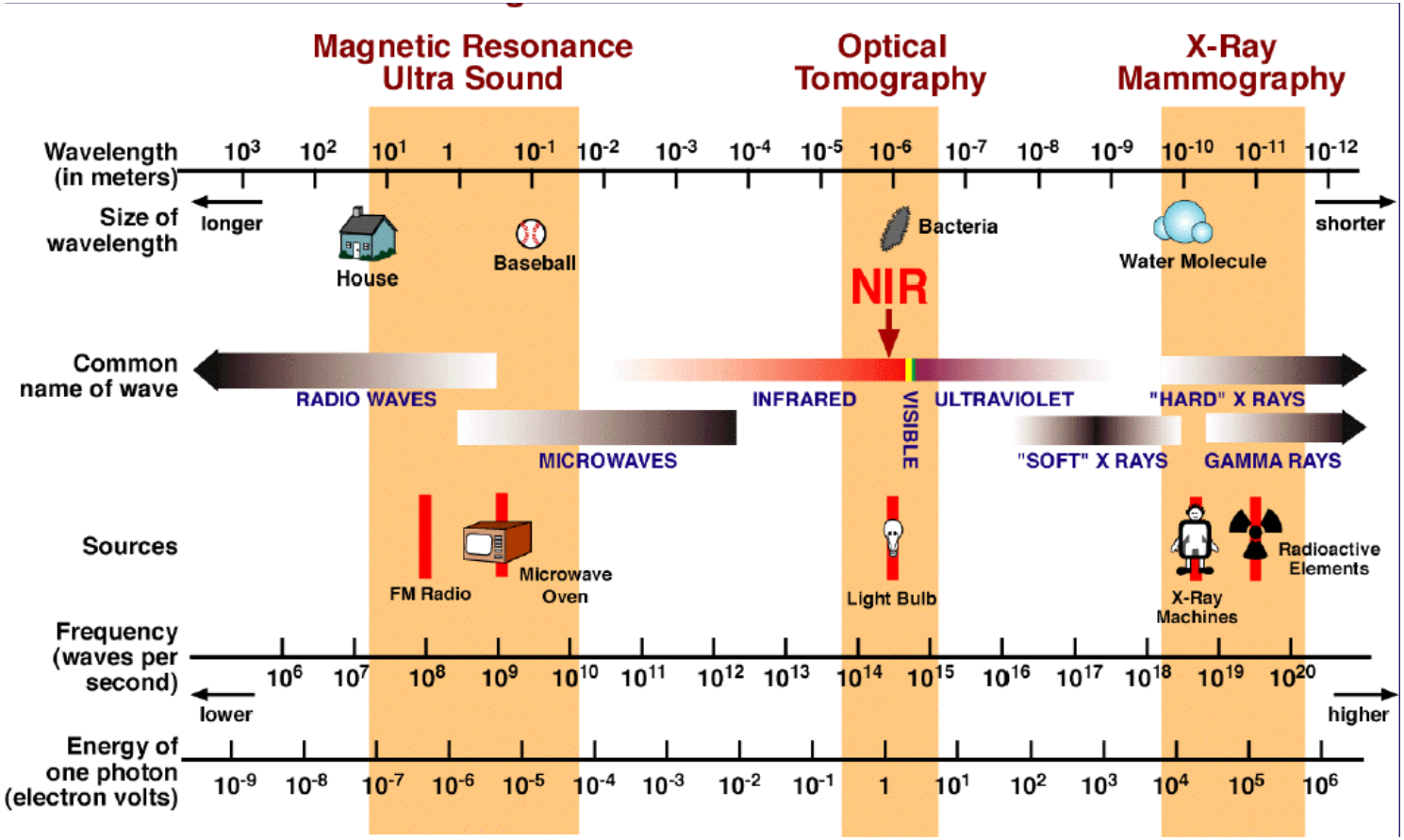
- 影像类：公司、高校
  - 1、GE、Siemens、Philips、东软、安科、联影、迈瑞……；2、清华、北大、西北大学、斯坦福……
  - 岗位：1、序列开发、算法、软件、现场工程师、临床需求；2、CT重建算法、功能成像、临床序列、快速成像……
- 放疗类：公司、高校、医院
  - 1、医科达（瓦里安+北医所）……；2、清华、北大、西北大学、斯坦福……；3、北肿瘤、中山医……
- 医学物理师（Medical Physicist）：在大学主修物理或其他科学科系，进而在研究所－物理所或医学物理所－取得硕士或博士，经至少两年住院医实习训练后取得资格。(AAPM: The American Association of Physicists in Medicine)

Medical Imaging System

- Introduce some form of radiation (energy)
  - Electromagnetic
  - Acoustic
- Observe its interaction with (human) tissue
  - attenuation
  - scattering / reflection
- Convert the observations into a clinically meaningful images
  - Film
  - Electronics\computer

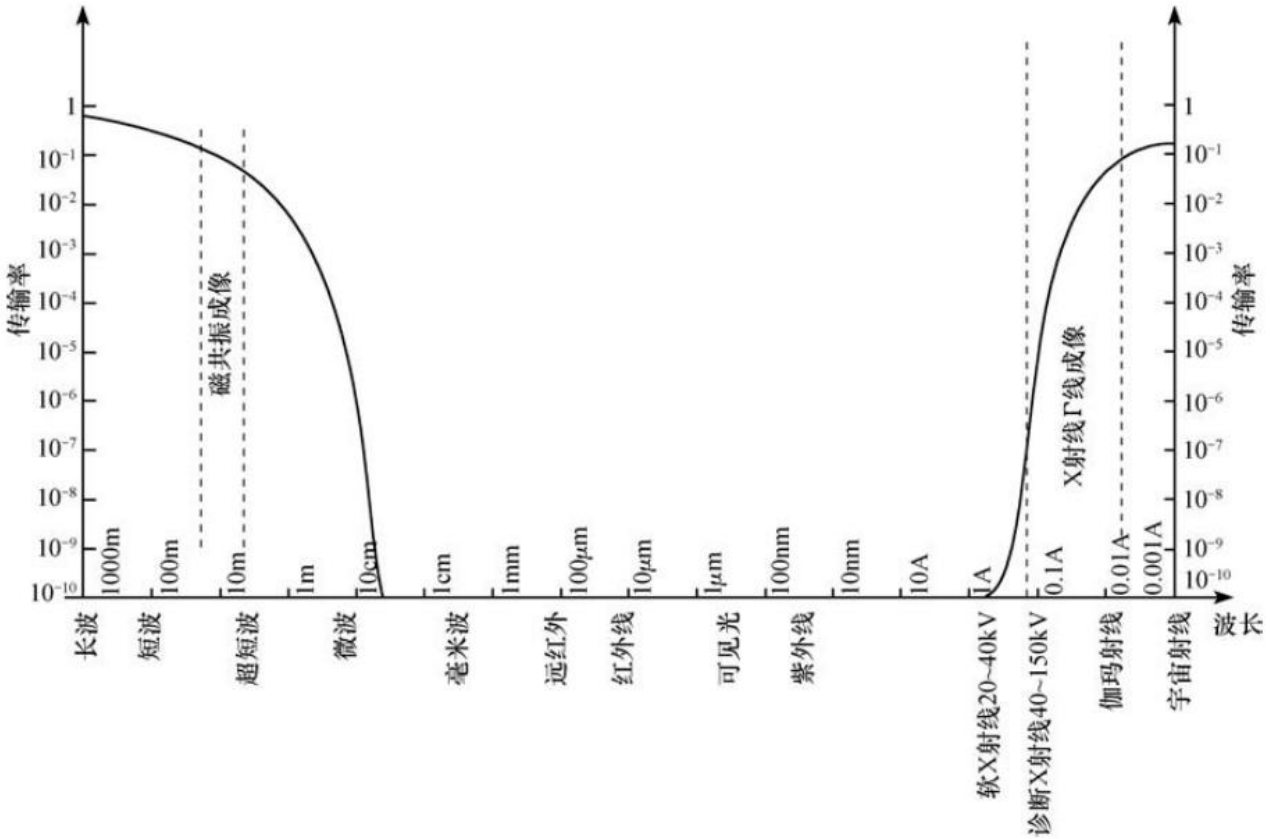
Different imaging methods have their own characteristics. Images from different sources carry different information and have their own applicability.

Electromagnetic Spectrum



$$E = h\nu = h \cdot \frac{c}{\lambda}$$

Transmission of electromagnetic waves through 25cm soft tissue



- Full attenuation, no transmission
- Full transmission, no attenuation
- Imaging conditions：
  - Shorter wavelength
  - Moderate attenuation

Imaging Considerations

- Type of information
  - Anatomical - from head to toe 结构(解剖)
  - Functional - cardiac, brain, etc. 功能
  - Qualitative vs. Quantitative 定性-定量
- Benefits \ Limitations
  - resolution
  - sensitive range (e.g. view angles)
  - speed
  - cost
  - invasiveness

- 光学显微成像
- 内窥镜
- 热像
- 。。。 。。

Invisible → Visible

“Classical” Imaging Methods

Images that are **direct** manifestations
of the interaction
between **radiation** and **tissue**

能量辐射与生物组织作用的直接表现

- Projection Radiography (Conventional X-ray)
- Conventional Ultrasonic Imaging
- Conventional Nuclear Medicine (eg. γ camera)

“Computed” Imaging Methods

Images that are formed using mathematical methods and computers from **indirect** measurements of the interaction between **radiation** and **tissue**

能量辐射与生物组织作用的非直接表现

- Computed Tomography (CT)
  - X-ray CT
  - PET
  - SPECT
- Magnetic Resonance Imaging
- 3D Ultrasound

Projection Radiography





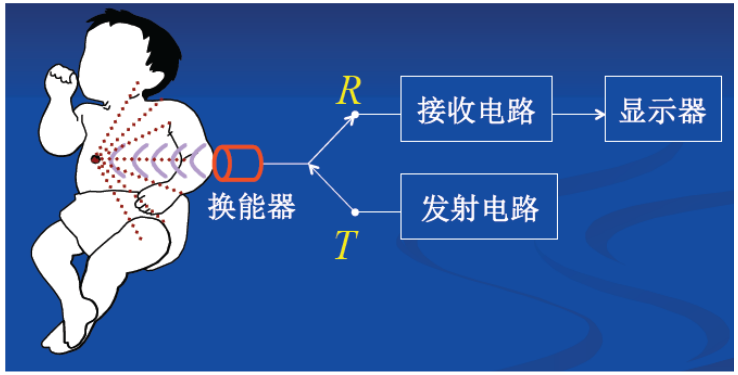
- **Physical Principle:**
  Variation in X-ray attenuation of different tissues
- **Methodology:**
  A beam of X-rays is directed through a patient onto a film.
- **Image type:**
  An X-ray "shadow" of the patient.
- **History:**
  Roentgen's discovery - 1895
  Application to medicine – 1896
  contrast materials - early 1900's
  angiography - 1927





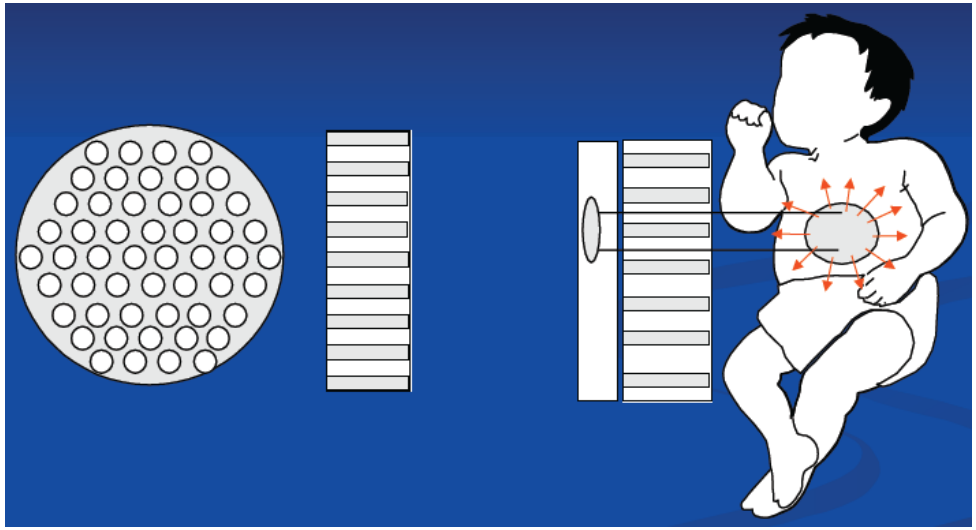
Ultrasonic Imaging

- **Physical Principle**:
  - Ultrasound waves scatter and reflect within the body
- **Methodology**:
  - A pulse of ultrasonic energy is propagated into the body and backscattered echoes record the depth of objects in the body.
- **Image type**:
  - A “depth map” of patient organs.
- **History**:
  - Concept derived from W.W.II sonar
  - Major clinical development - 1970’s



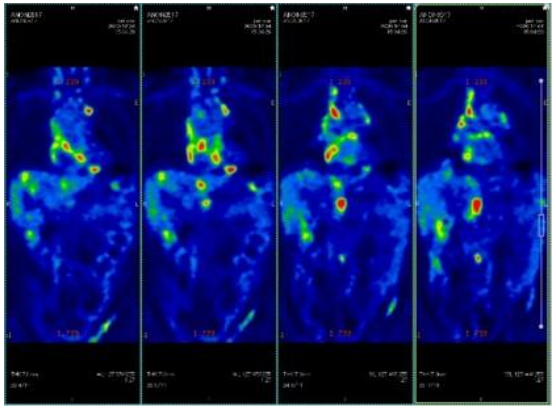
Conventional Nuclear Medicine

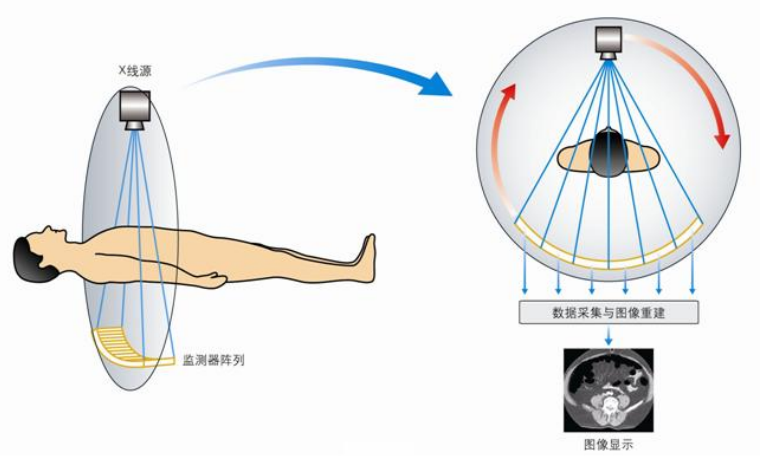
- **Physical Principle**:
  - Variable uptake of radioactive materials by different organs
- **Methodology**:
  - Inject patient with radio-labeled substance and record time space pattern of radiation.
- **Image type**:
  - A map of the radioactivity of the patient.
- **History**:
  - Therapeutic administration of radio-labeled substances - 1950
  - Scintillation camera - 1952



Computed Tomography

- **Physical Principle**:
  - Projection Slice Theorem dictates how to reconstruct a 2-D image from multiple 1-D projections (Radon Transform).
- **Methodology**:
  - Obtain multiple projection images and reconstruct images using a computer.
- **Image type**:
  - A 2-D slice mapping the patient's X-ray attenuation coefficient (X-ray CT) or radioactivity (PET and SPECT).
- **History**:
  - X-ray CT proposed - mid 1960's
  - Early clinical use - 1972
  - PET and SPECT followed X-ray CT





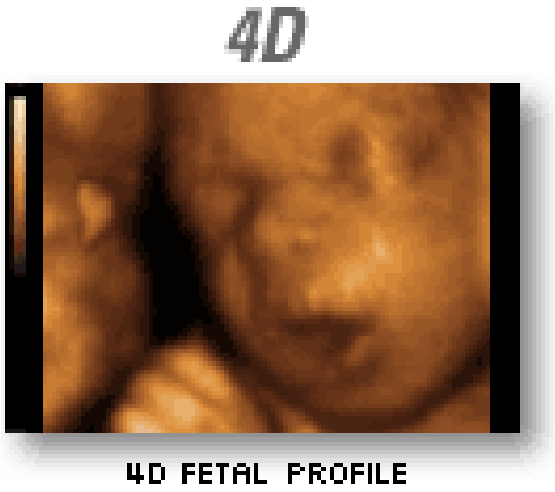
$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Magnetic Resonance Imaging

- **Physical Principle**:
  - Within a strong magnetic field, paramagnetic nuclei (usually hydrogen protons) will Resonate in response to RF radiation
- **Methodology**:
  - Place patient in a magnet, irradiate with RF field, and record spatially encoded RF echoes.
- **Image type**:
  - A map of proton concentration through a slice of the body.
- **History**:
  - NMR discovered - 1940's
  - Imaging proposed in 1972
  - Current generation of machines developed in 1980's

The future

- Multi - Dimensional
- Multi - Parameters
- Multi - Modalities
- Anatomic - Functional
- Macro-Micro
- Manual-Automatic
- Picture Archiving and Communication System (PACS)



4D FETAL PROFILE

Important issues in Medical Imaging system

- Hazard
- Resolution
- Parameters
- Properties of radiation
- Regions of imaging
- field of view
- Image noise
- Imaging speed
- Price of the equipments
- cost of diagnosis
- ……

X线：电离辐射
X－CT：电离辐射
超声：安全
核素：同位素照射
MRI：安全

X线：X线衰减系数
X－CT：X线衰减系数
超声：反射/散射/运动
核素：同位素衰变
MRI：T1/T2/pd

X线：透视/拍片
X－CT：数秒/数分钟
超声：实时
核素：数秒/数分钟
MRI：数秒/数分钟

X线：≈1mm
X－CT≈1mm
超声：0.3~3.0mm
核素：≈ 10mm
MRI：≈1mm

X线：大
X-CT：大
超声：声窗限制
核素：大
MRI：大

X线:低/中
X-CT：中/高
超声：低/中
核素： 中/甚高
(PET)
MRI：甚高

Origins of Digital Image Processing

“高性能”计算机
(1960s)
(Storage,Display)

太空计划
(1960s)
(漫步者7号)

放射医学
(late 1960s,
early 1970s)

Internet
（1990s）

Digital Image Processing

图像增强
(Image Enhancement)

图像恢复
(Image Restoration)

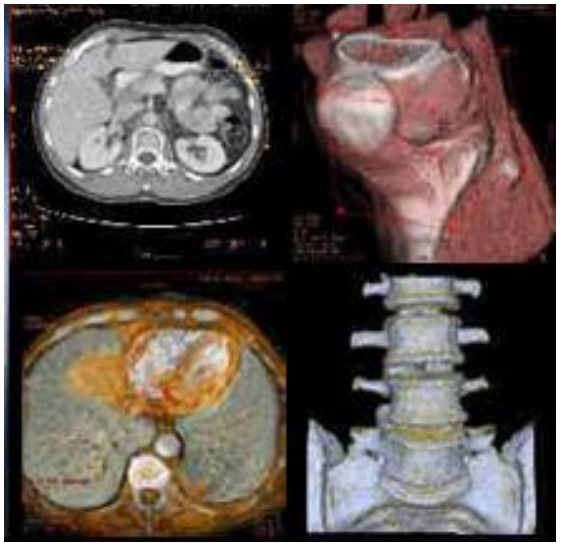
图像重建
(Image Reconstruction)

图像分割
(Image Segmentation)

图像压缩
(Image Compression)

图像传输
Image Transmission







Introduction

1. 什么是数字图像处理

   What is digital image processing?

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   The origins of digital image

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   Examples of fields that use digital image processing

4. 数字图像处理的基本步骤

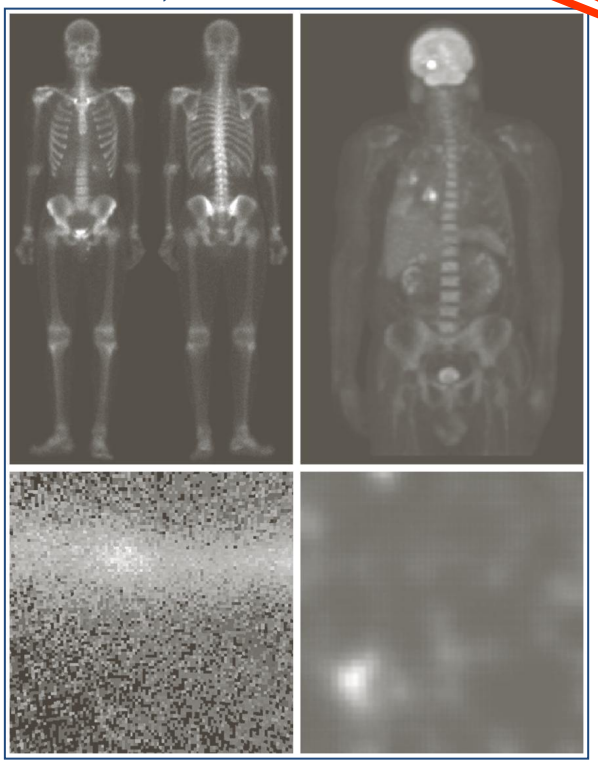
   Fundamental steps in digital image processing

5. 数字图像处理系统组成

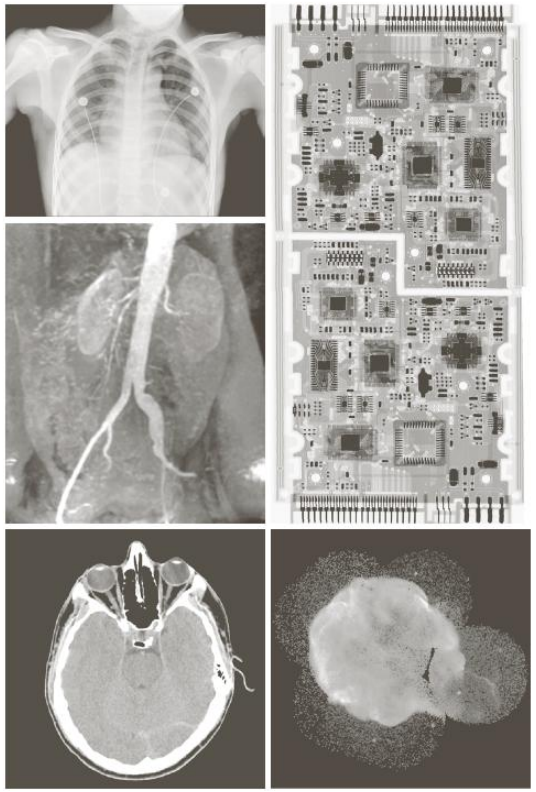
   Components of an image processing system

Examples of fields that use DIP



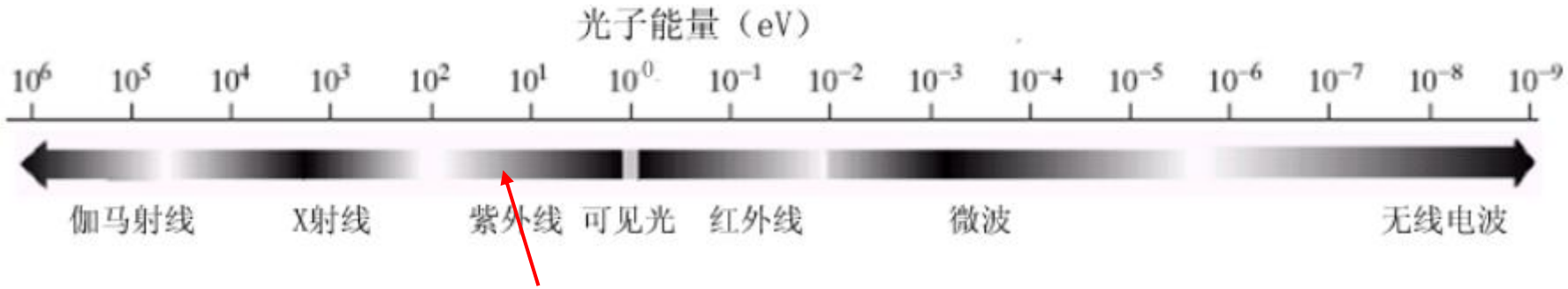


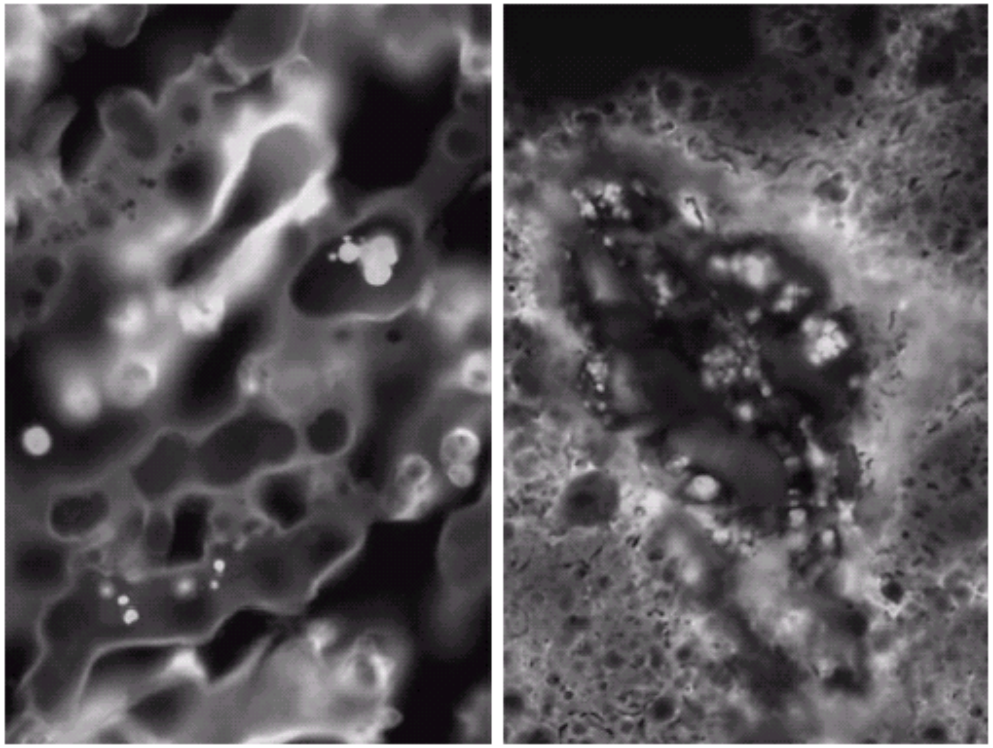
Gamma imaging, mainly used in medicine and astronomy



X-ray imaging: Mainly used in medicine (chest radiograph, angiography, tomography) and industrial inspection

Examples of fields that use DIP





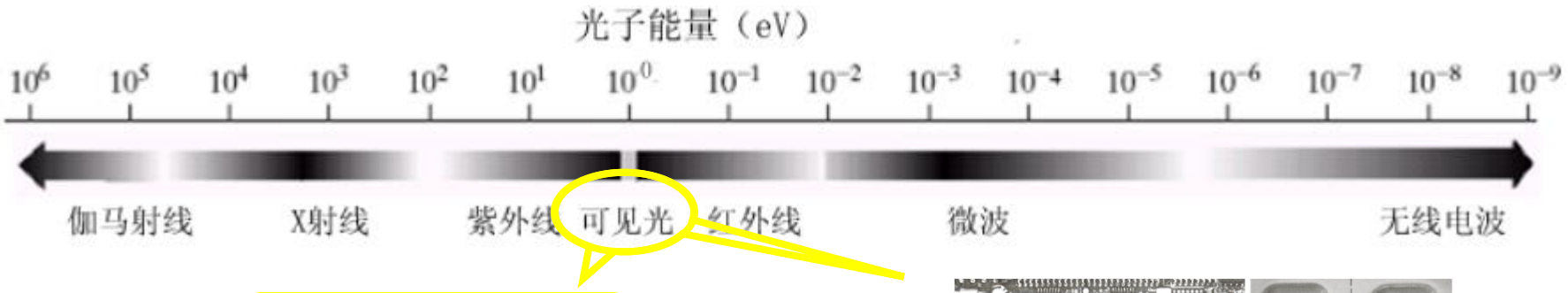
普通玉米

患黑穗病玉米

Ultraviolet band imaging：Minerals, grains, textiles

Fluorescence microscopy
荧光显微技术

Examples of fields that use DIP



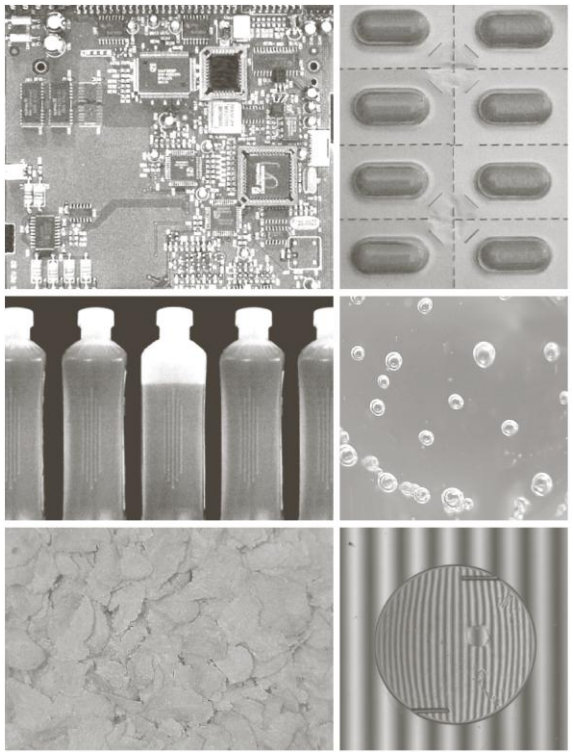


指纹识别

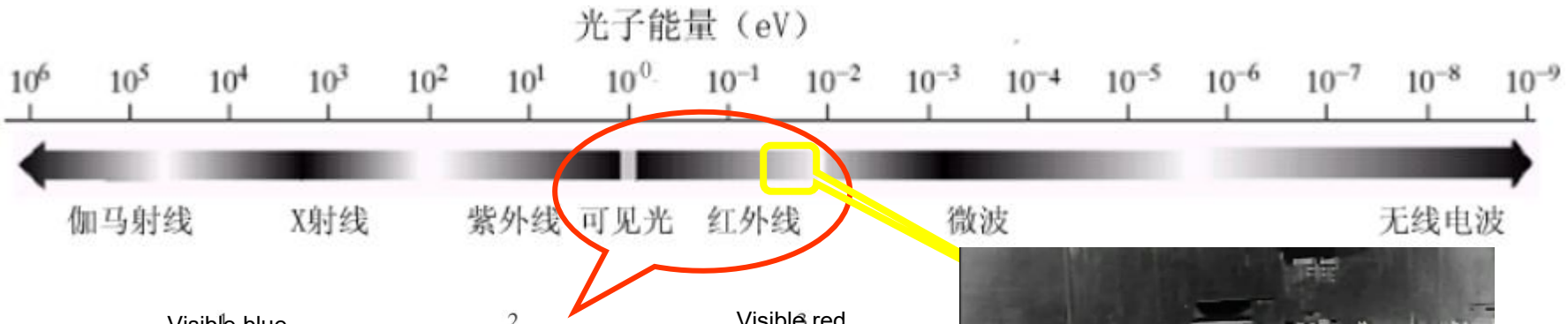
纸币识别



车牌识别

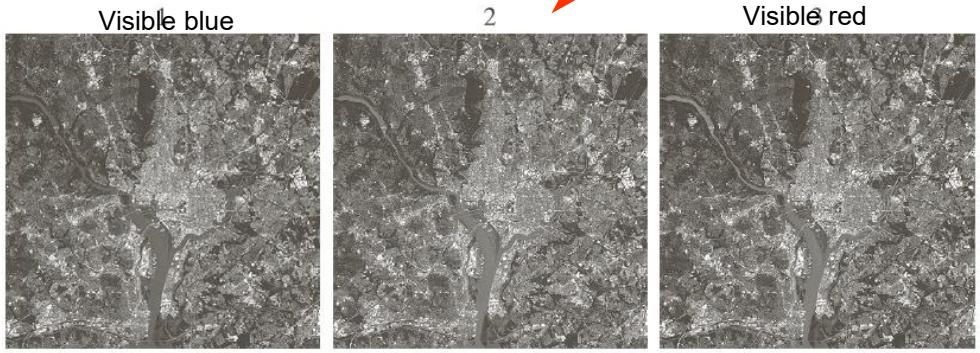


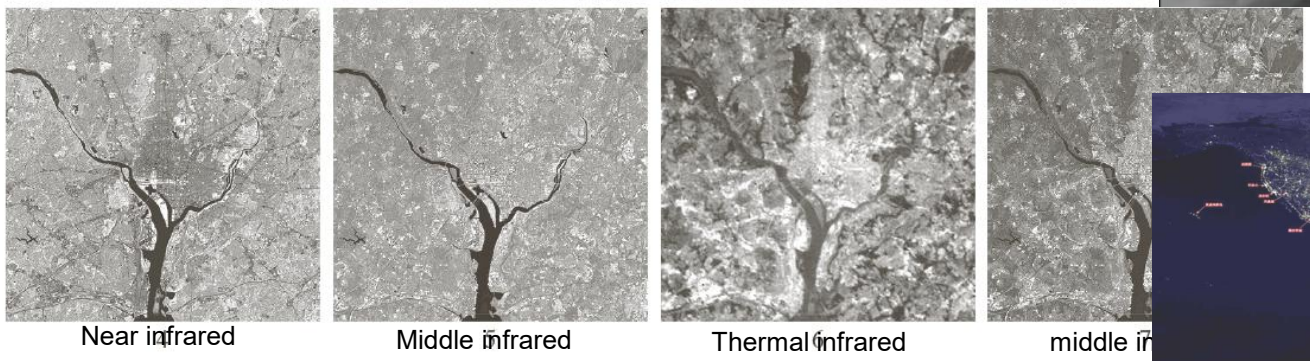
Examples of fields that use DIP





多光谱遥感

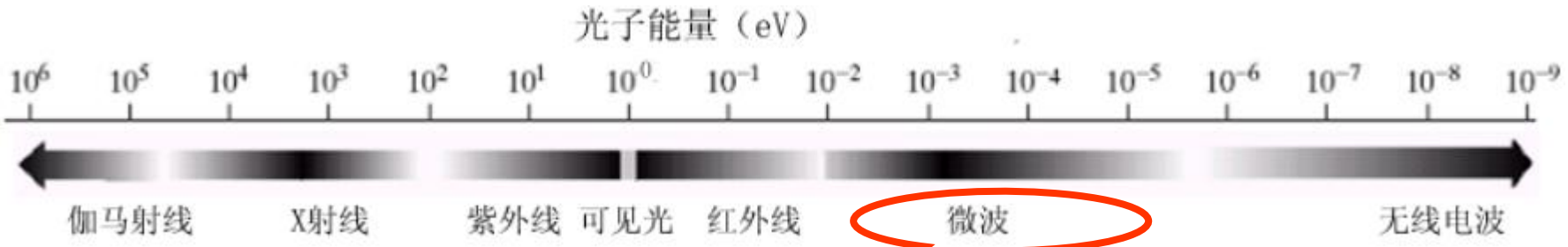






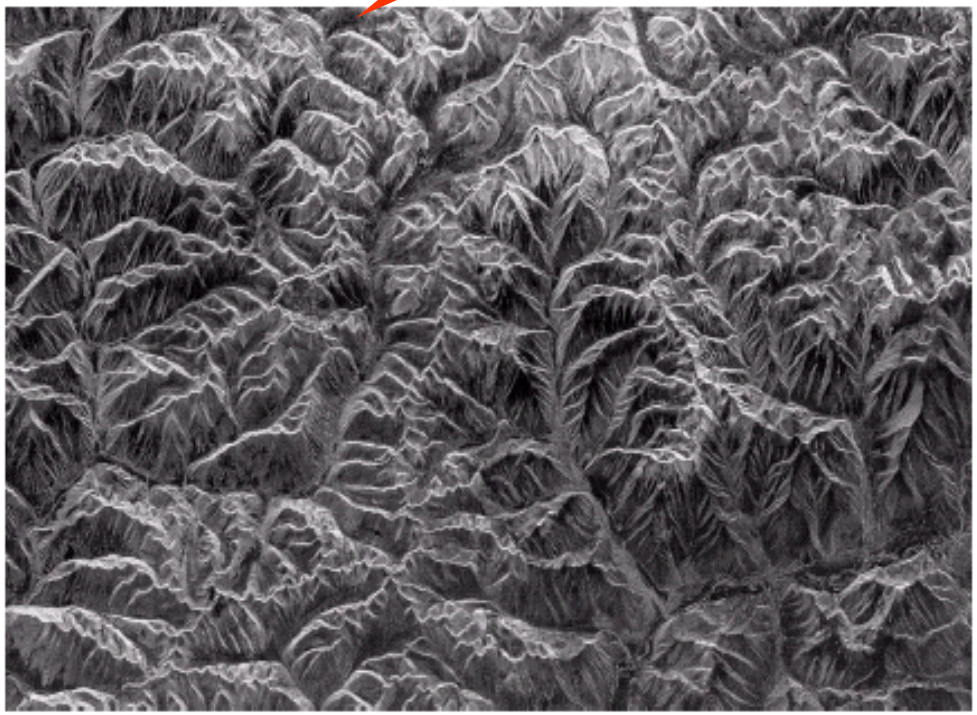
全球夜景卫星红外成像

Examples of fields that use DIP

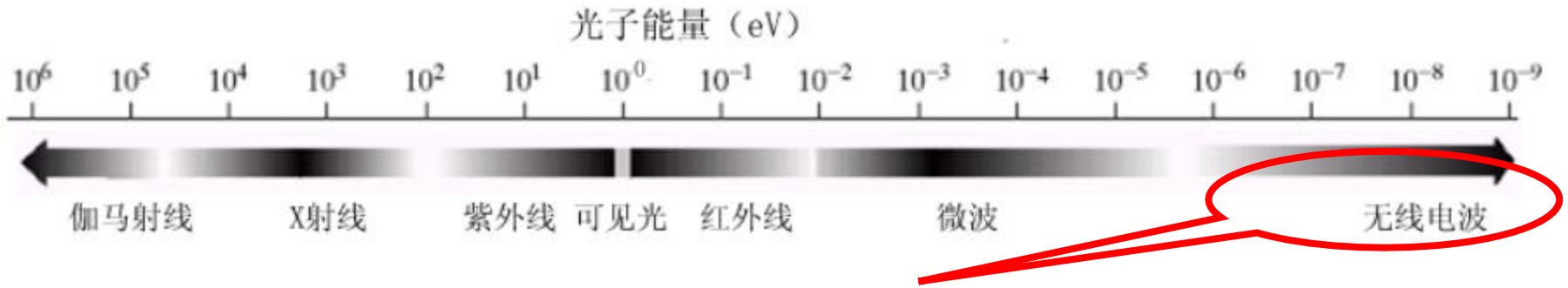


西藏东南部高低不平山区的星载雷达图像(图像由NASA提供)

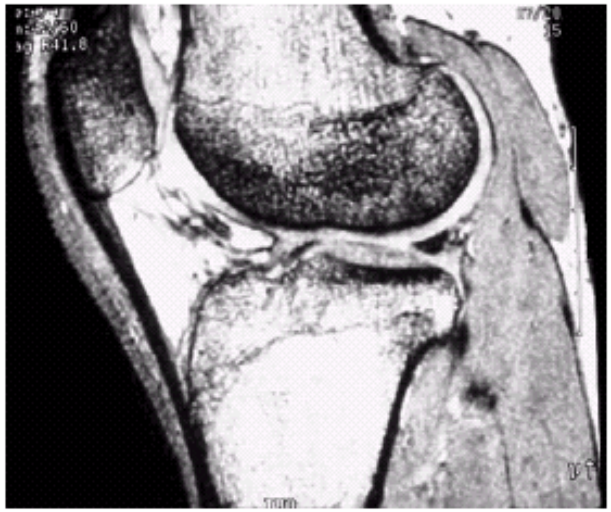
图像清晰度和细节未被云层和其他大气条件阻挡，通常这些因素在可见光波段会影响图像



Examples of fields that use DIP



Magnetic Resonance Imaging (MRI, 核磁共振成像，1980s)





1985年我国引进第一台
2003年 Nobel Prize

Introduction

1. 什么是数字图像处理

   What is digital image processing?

2. 数字图像处理的起源

   The origins of digital image

3. 使用数字图像处理领域的实例

   Examples of fields that use digital image processing

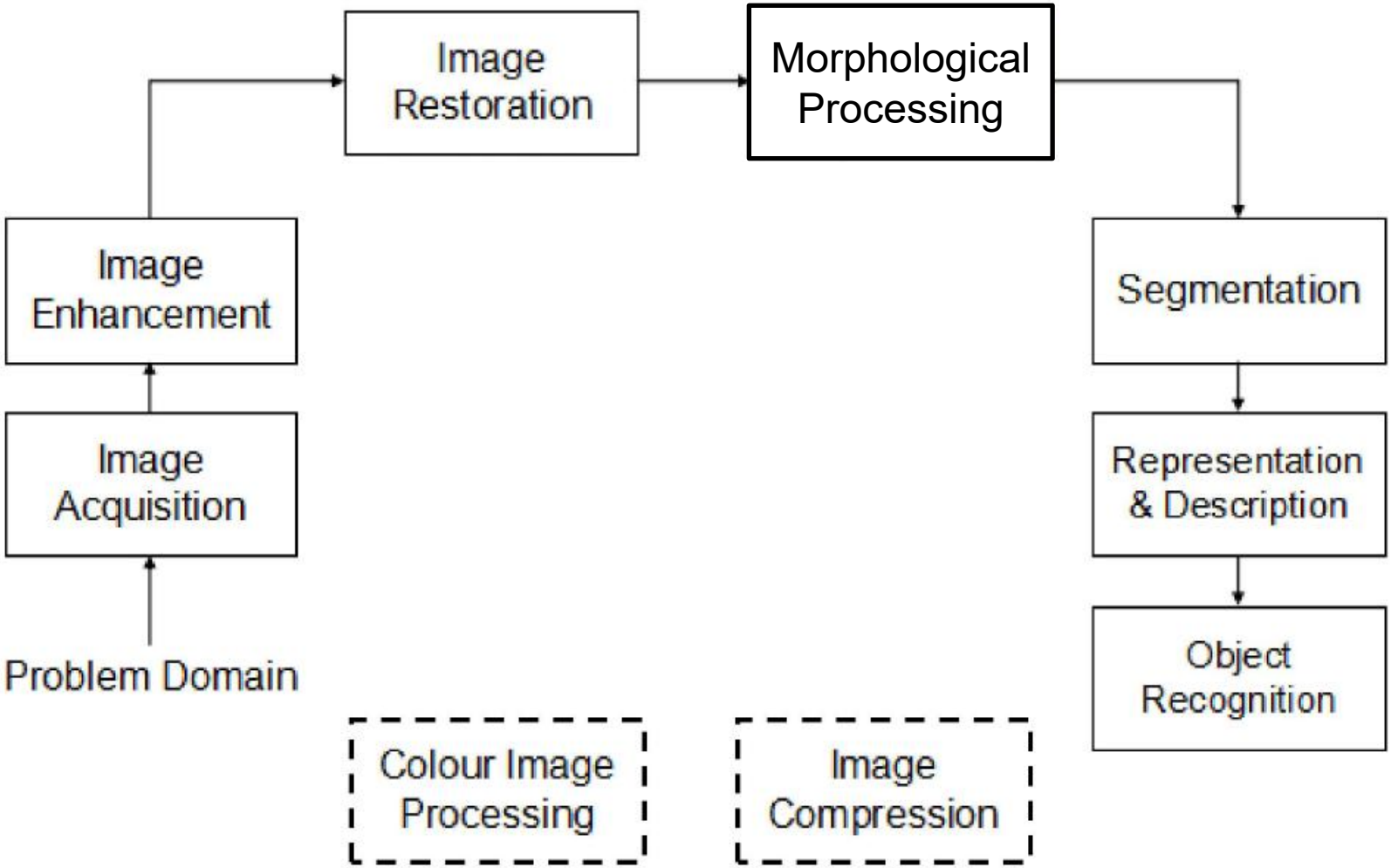
4. 数字图像处理的基本步骤

   Fundamental steps in digital image processing

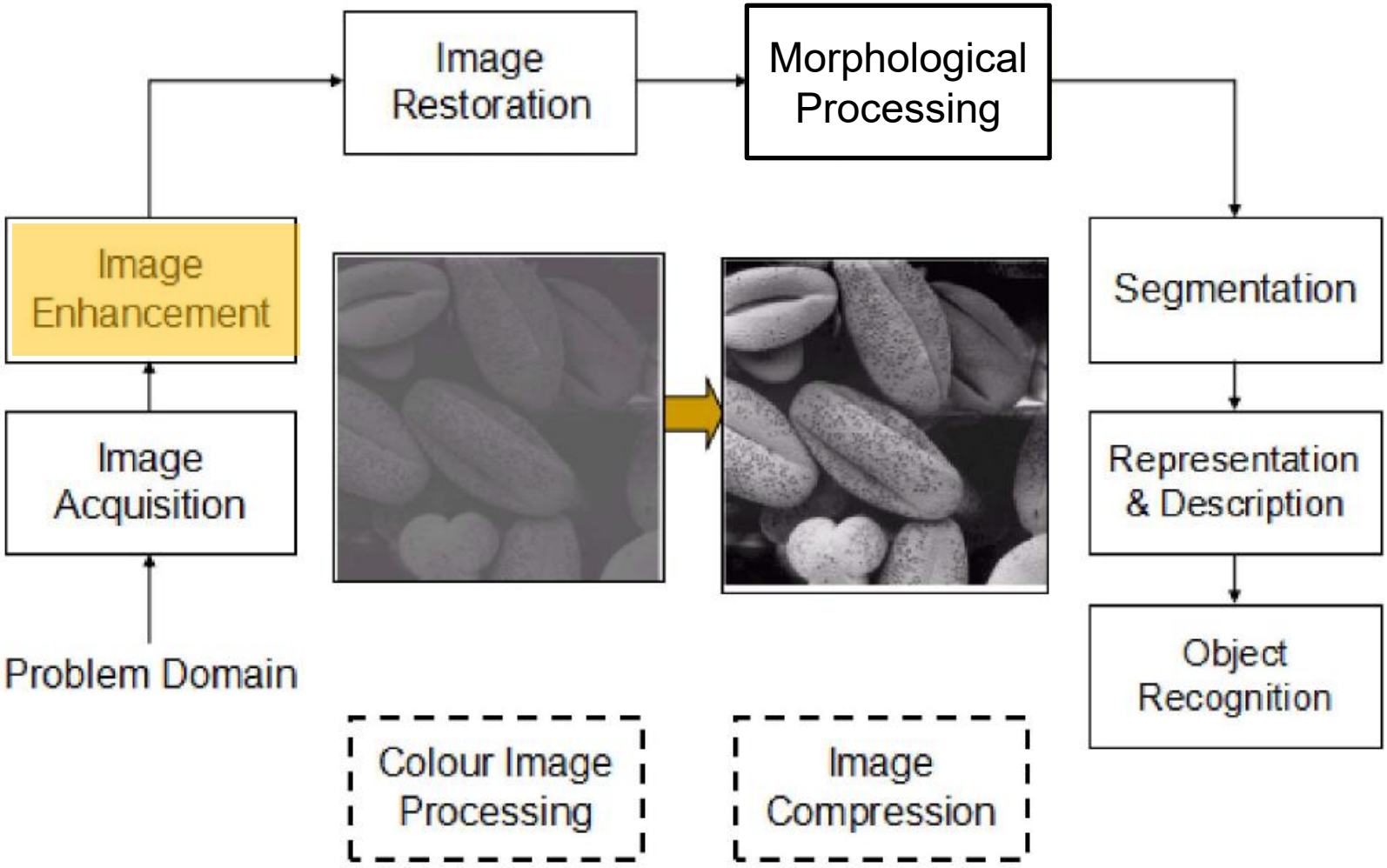
5. 数字图像处理系统组成

   Components of an image processing system

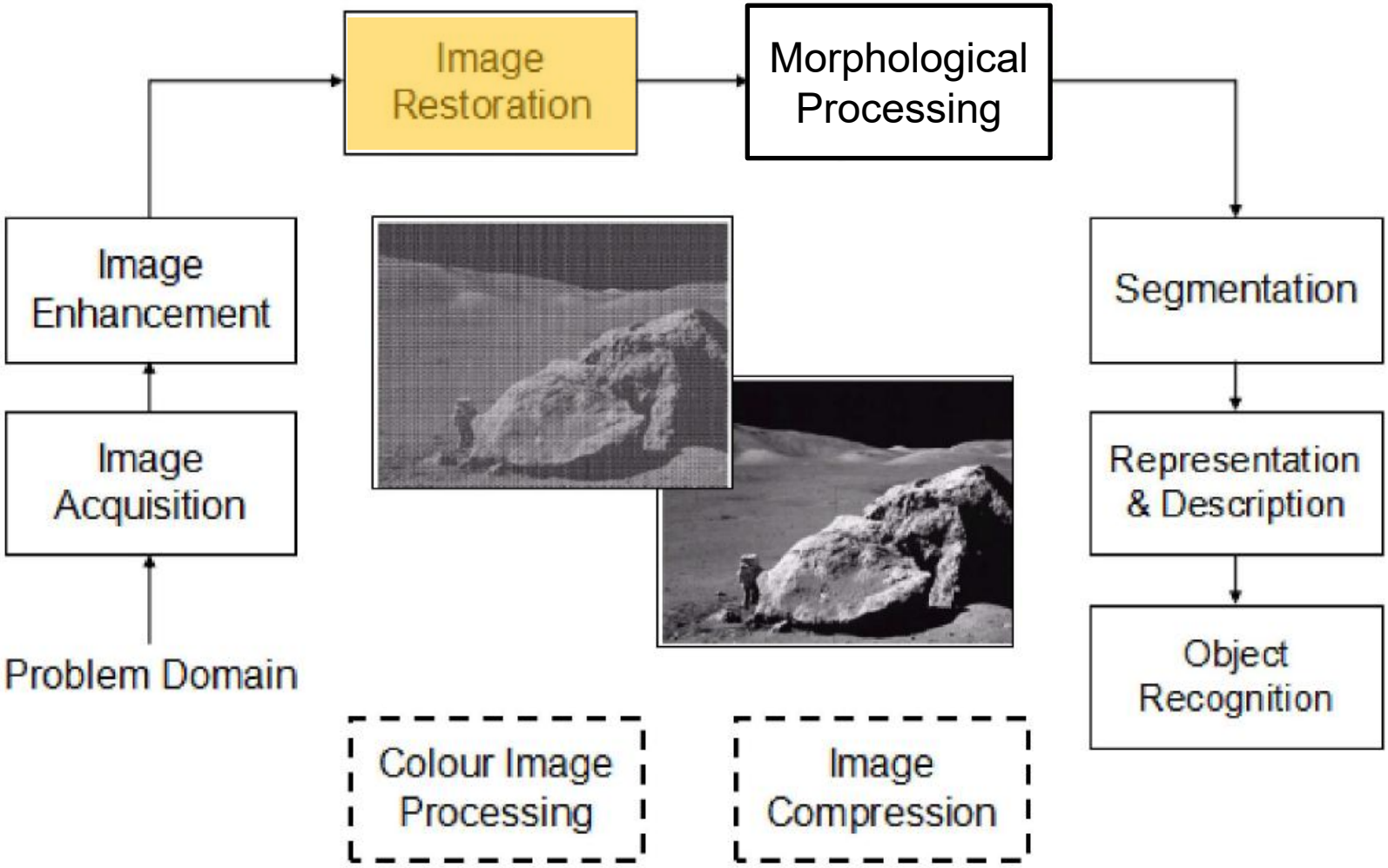
Fundamental Steps in DIP



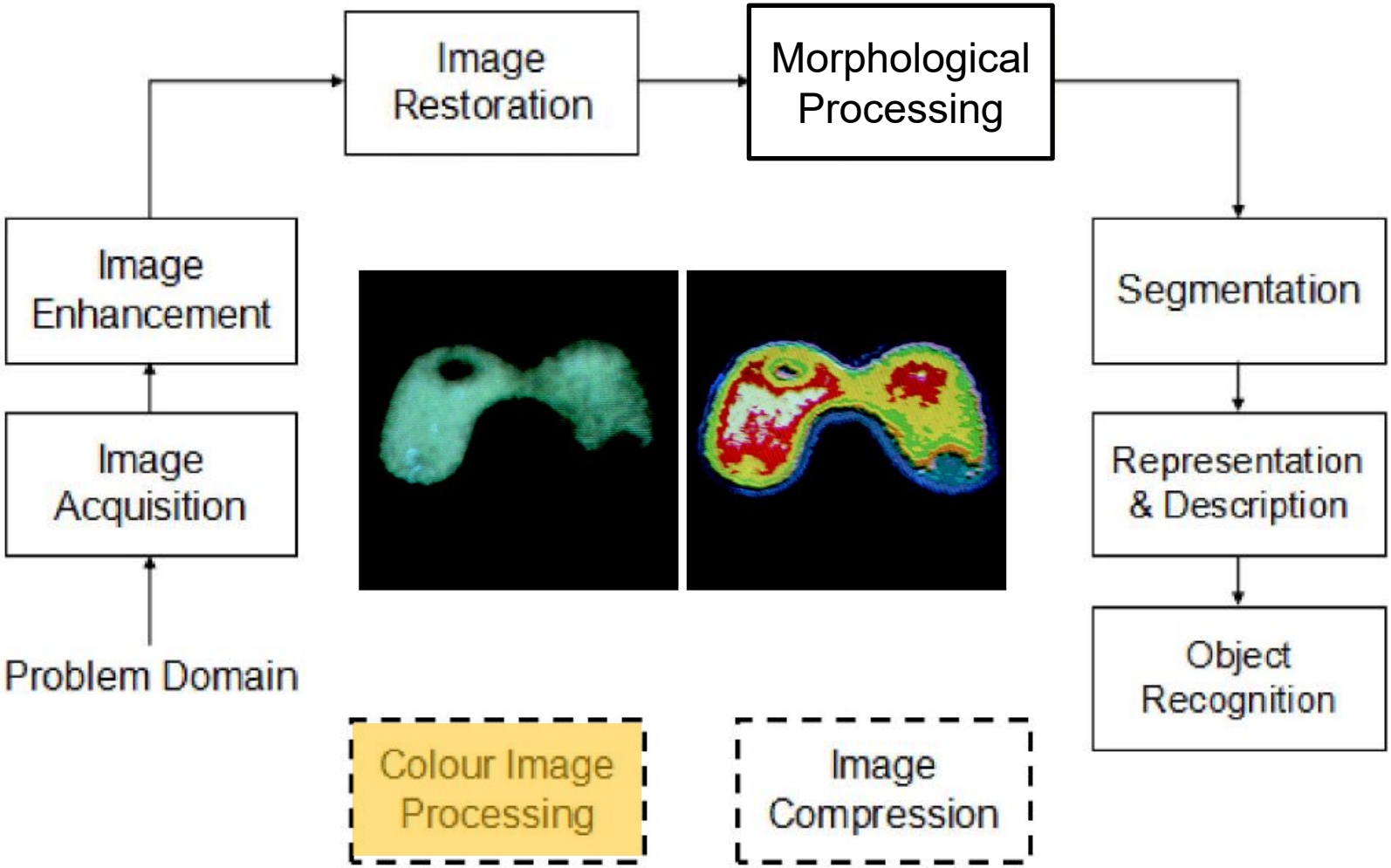
Fundamental Steps in DIP



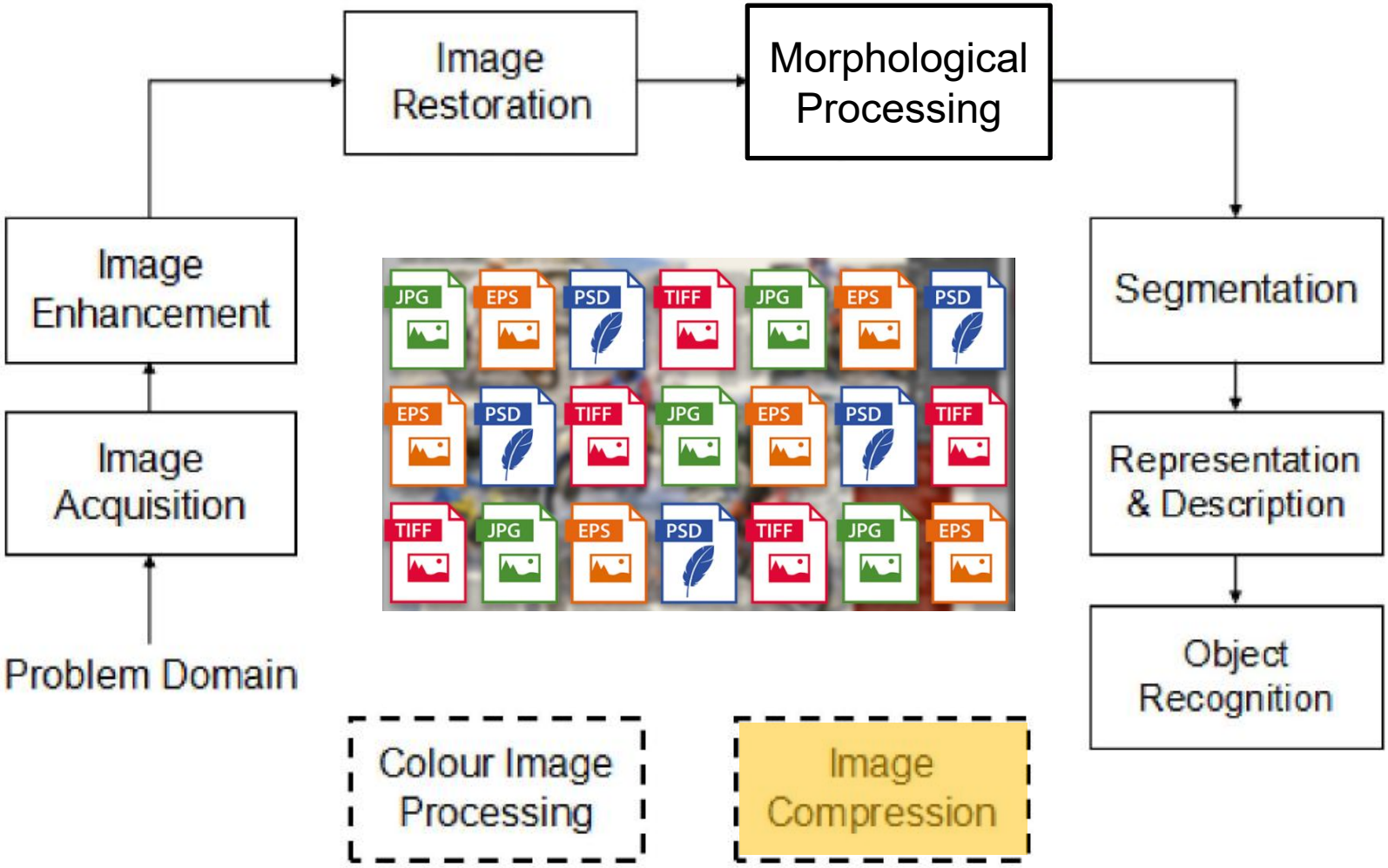
Fundamental Steps in DIP



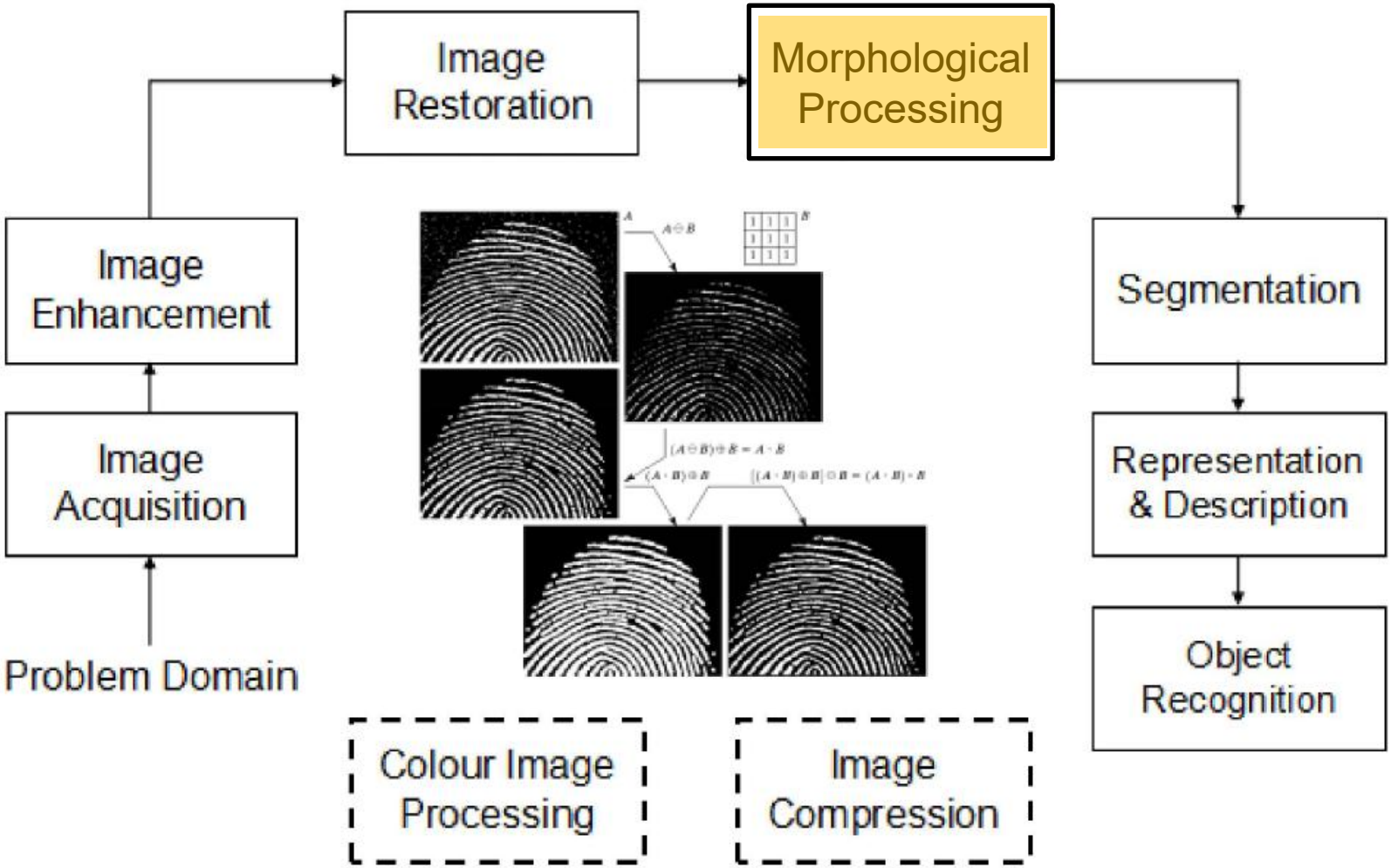
Fundamental Steps in DIP



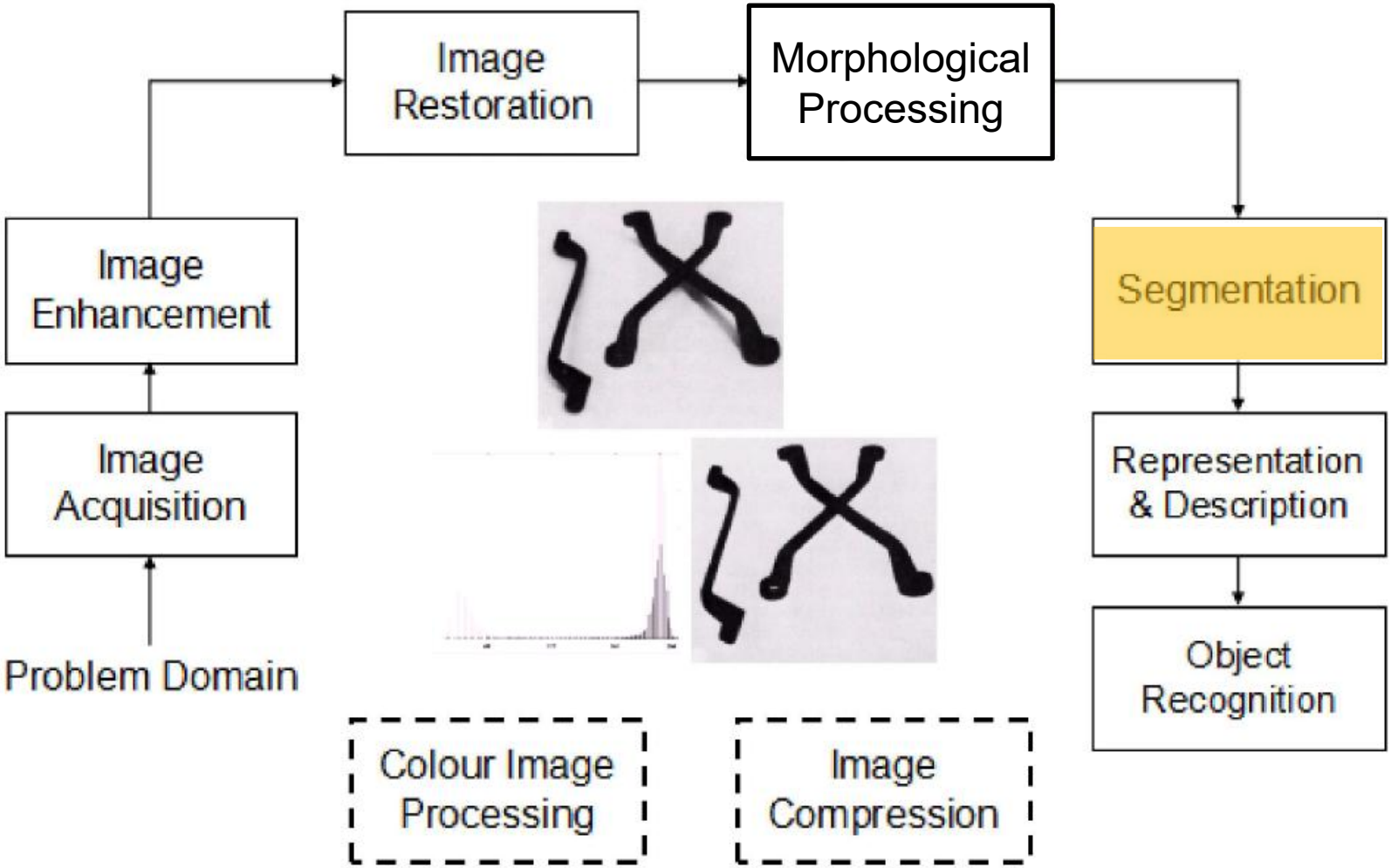
Fundamental Steps in DIP



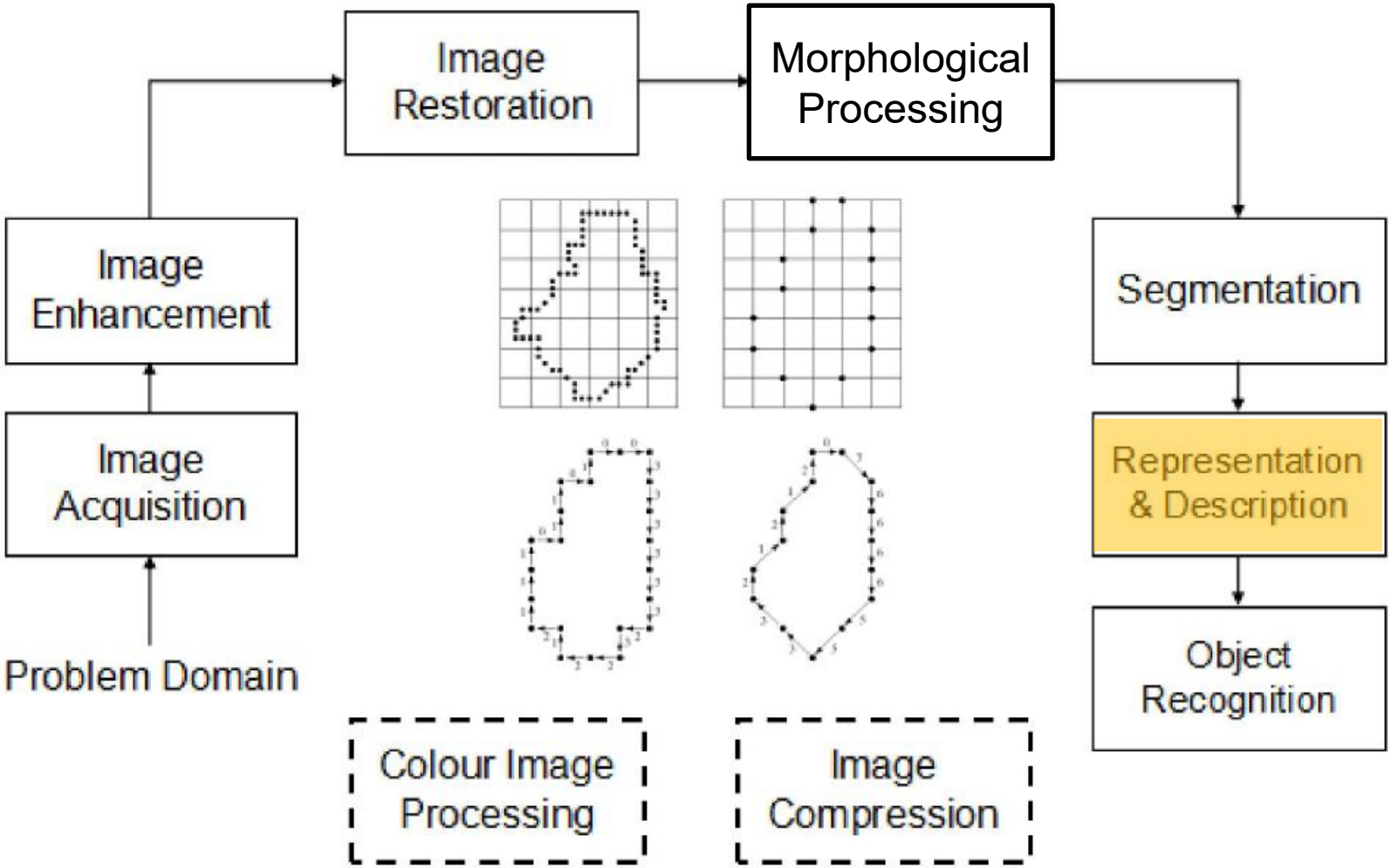
Fundamental Steps in DIP



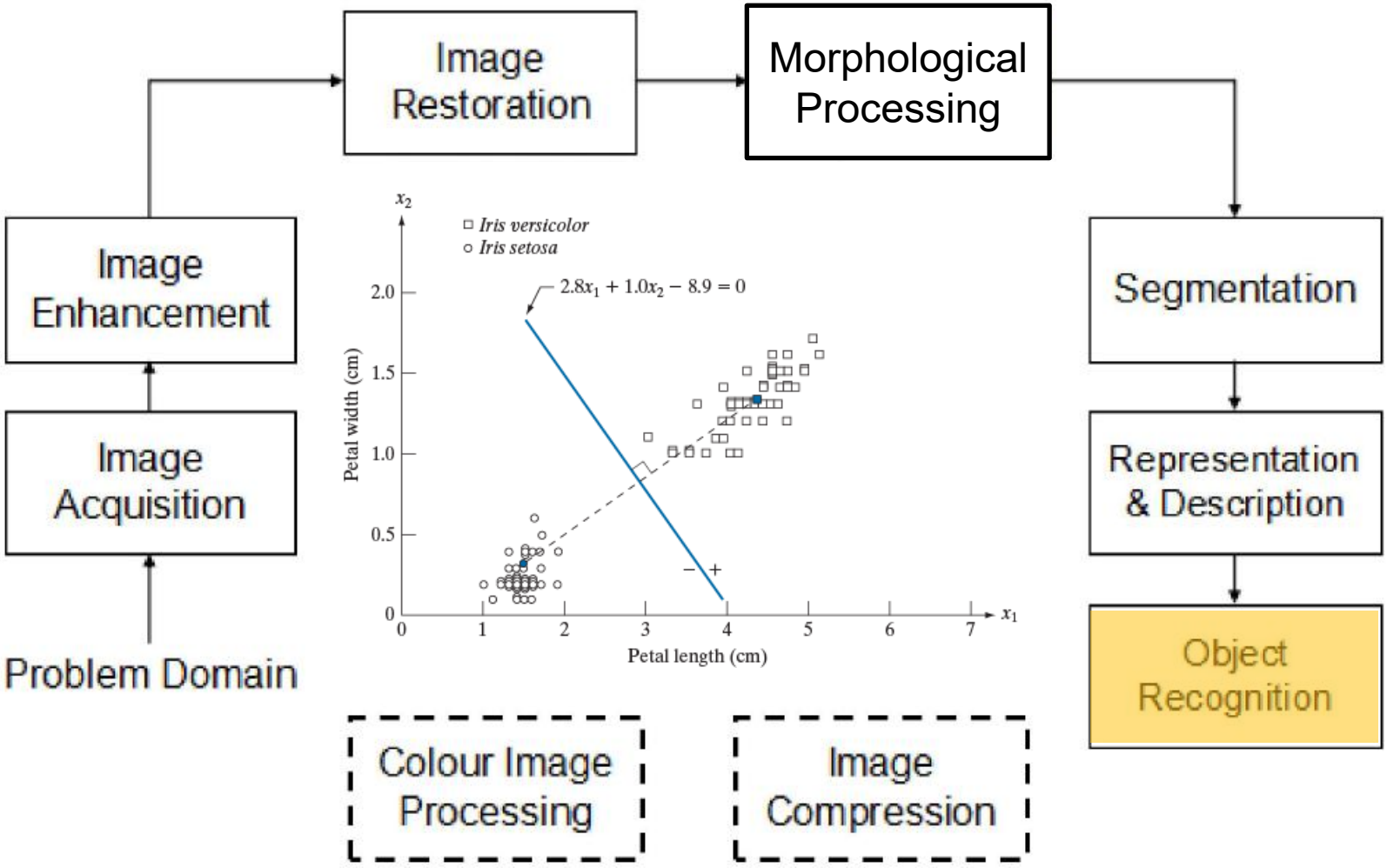
Fundamental Steps in DIP



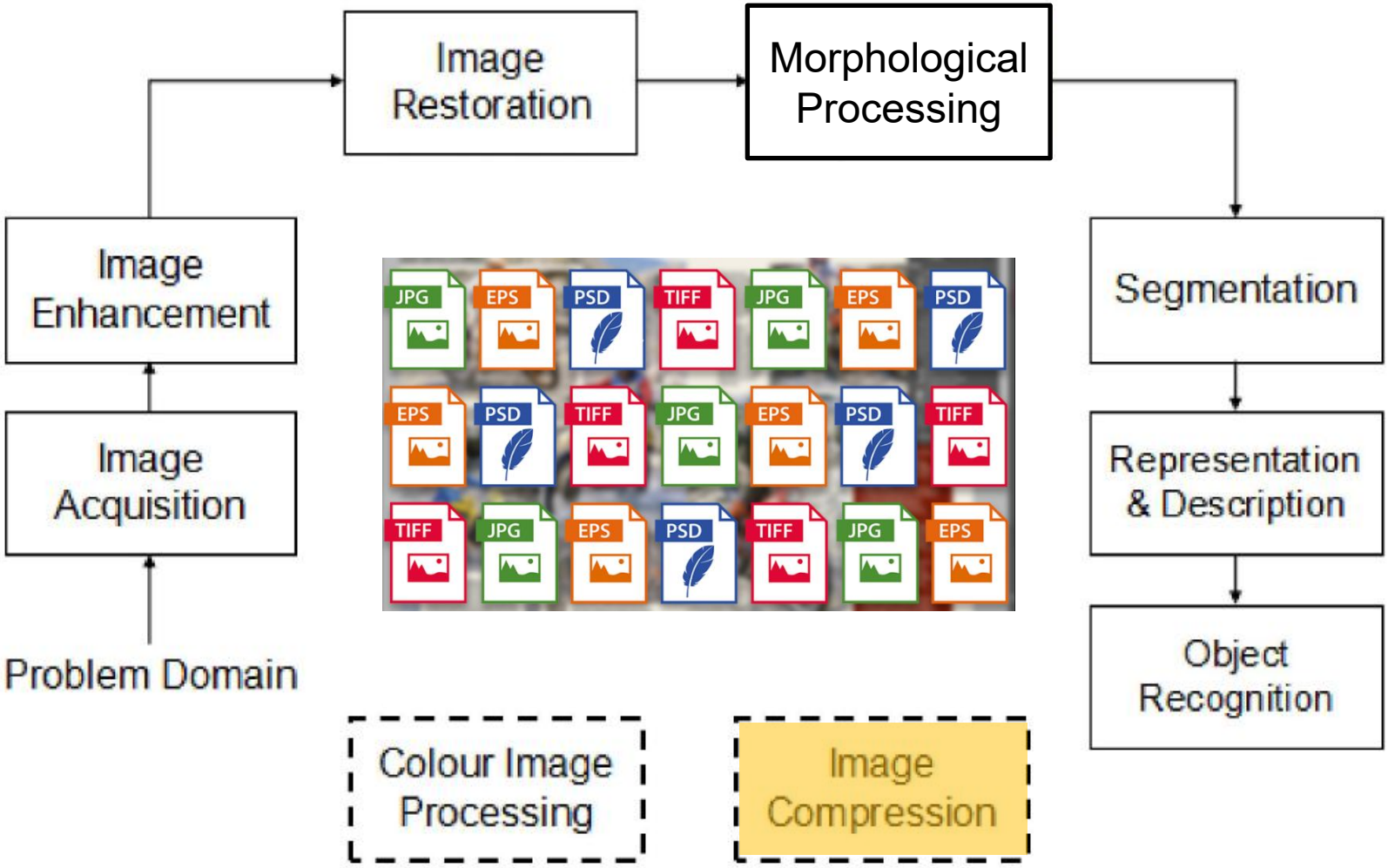
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Fundamental Steps in DIP



Fundamental Steps in DIP





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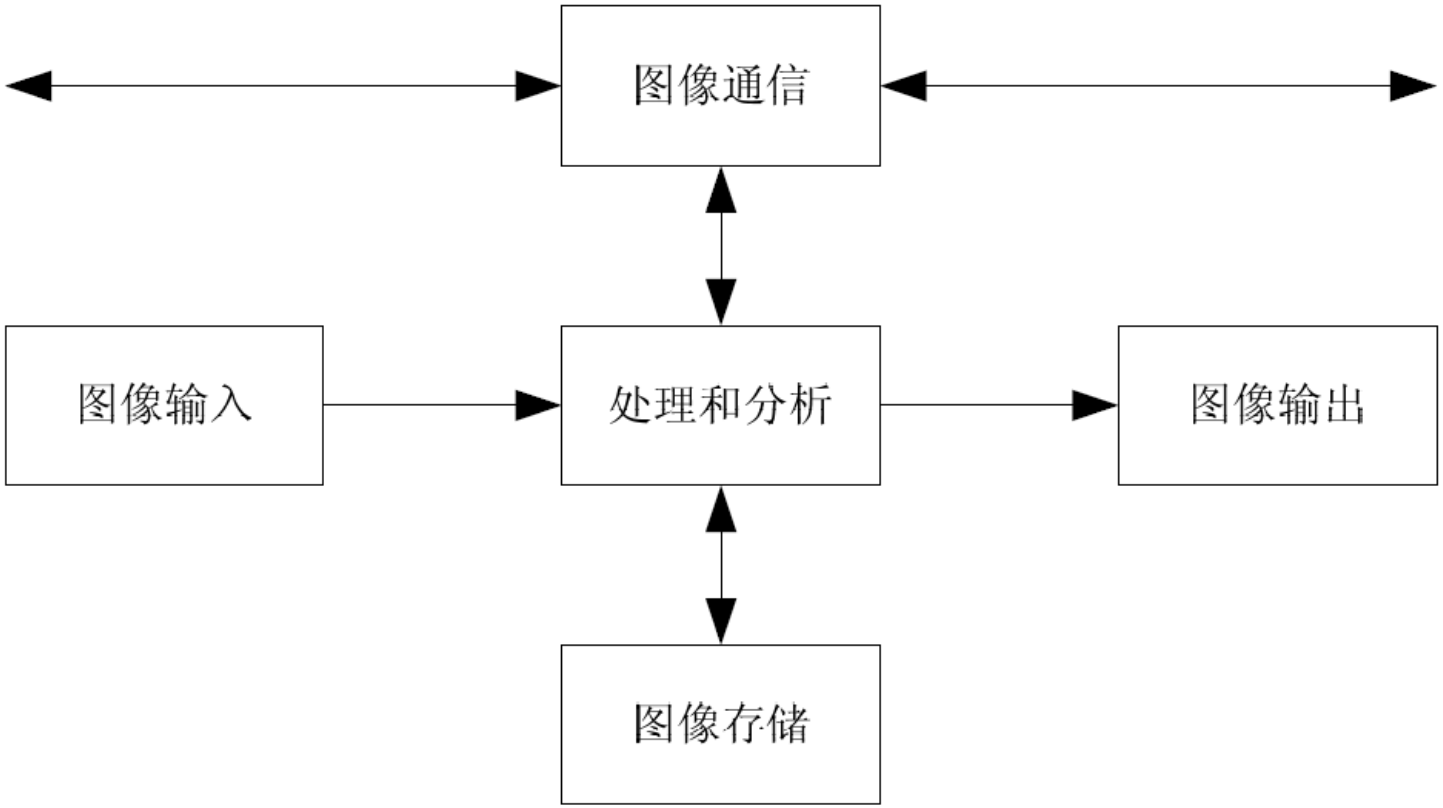
   Fundamental steps in digital image processing

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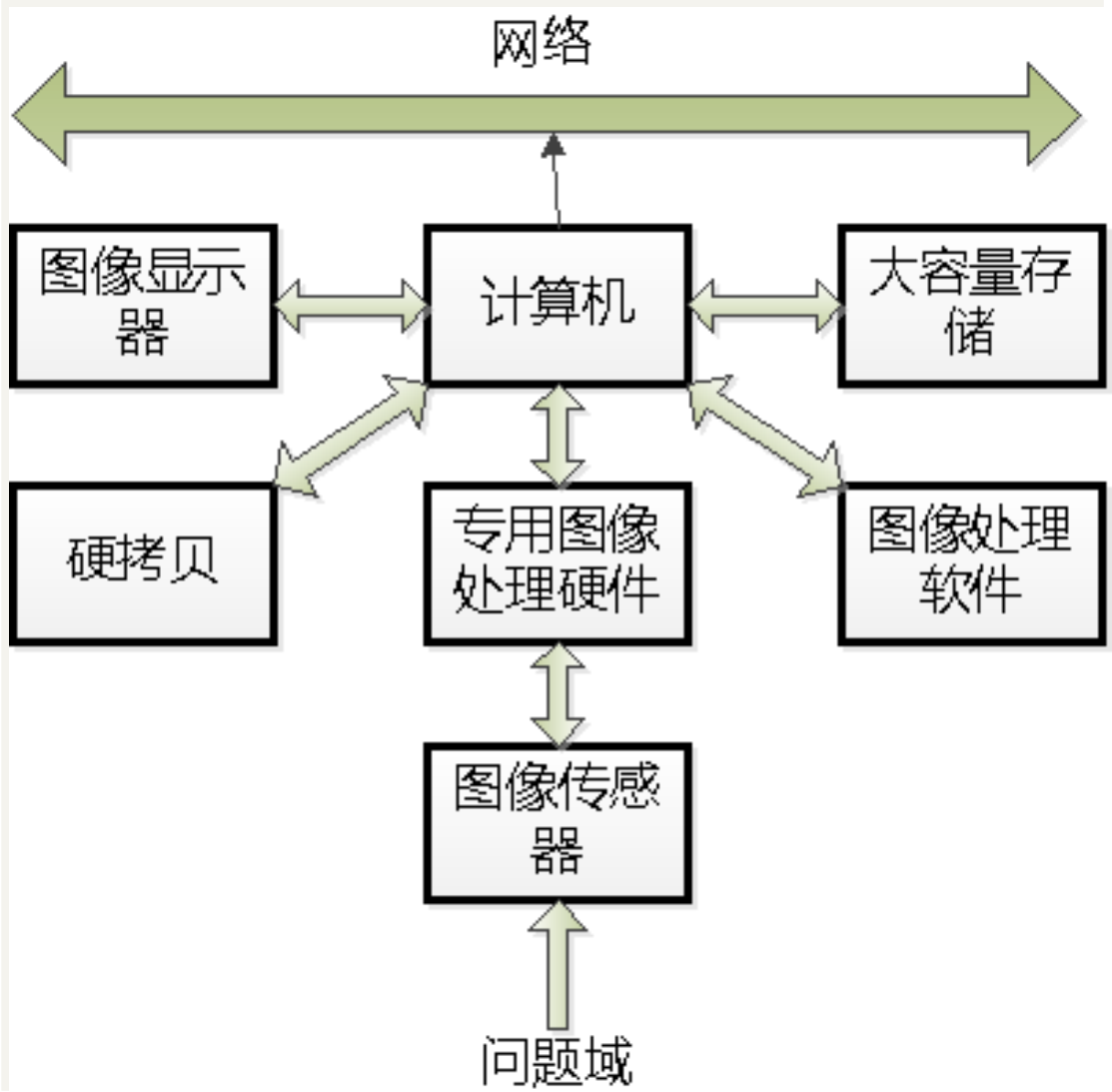
   Components of an image processing system

Components of an image processing system

- **It is composed of five modules: image input, image storage, image communication, image processing and analysis, and image output.**

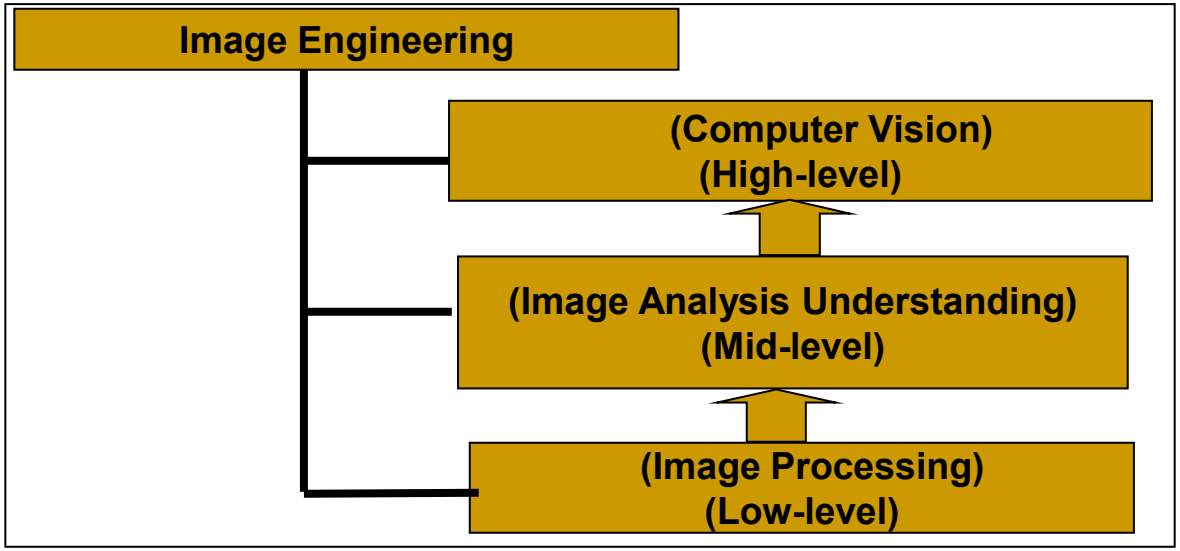


Components of an image processing system



Summary

- 数字图像处理
  - 什么是图像
  - 什么数字图像
  - 什么是数字图像处理



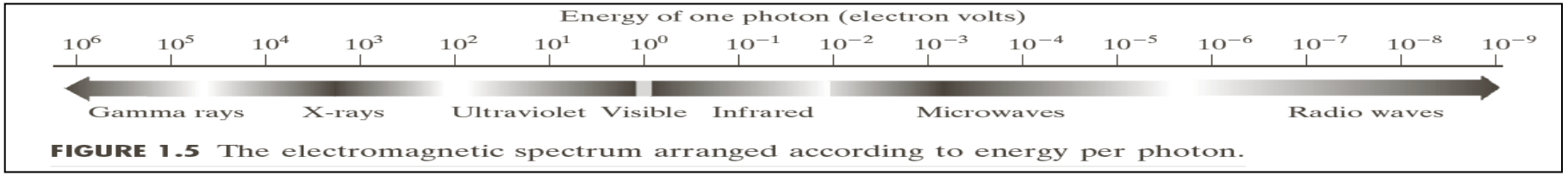


FIGURE 1.5 The electromagnetic spectrum arranged according to energy per photon.

